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"It really just wanted to talk about whales"

Addressing Misconceptions of LLMs amongst Lay Audiences

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Abstract

Artificial Intelligence (AI) has evolved through repeated cycles of excitement and disappointment, but the recent rise of Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs) has sparked unprecedented public interest. This raises the question of whether the general public has a sufficient understanding of the capabilities and limitations of the technology, and if not, how we can help them get there.

This thesis addresses the knowledge gap by presenting a taxonomy of misconceptions surrounding LLMs, based on available literature. We also present a mediation tool that is designed to help users reach a more accurate understanding of the technology through an interactive learning experience where users can build their own Large Language Model (LLM) by selecting different features in a skill-tree, which are connected to scrollytelling chapters that explain the features in more detail. Finally, we present a study on the prevalence of misconceptions in the different dimensions of the taxonomy, and the effectiveness of the mediation tool in resolving these misconceptions. The study consisted of 10 participants, who were interviewed before and after using the mediation tool. The interviews were coded based on the taxonomy, and the changes in misconceptions were analyzed.

The results show that a majority of the participants in the study had misconceptions such as anthropomorphization, real-time information access, and described use that could be categorized as cognitive passivity, though environmental impact appeared to be well understood. The mediation tool was found to be effective in addressing misconceptions like anthropomorphization, hallucination blindness and communicating the tokenization process, but work can still be done in addressing training data and context window misconceptions.

Acknowledgements

Looking back on my years at university, it is amusing to think that an early fascination with the Enigma machine eventually led me here. What began as an interest in mathematics has, somewhat unexpectedly, culminated in a thesis on the linguistic nature of the imitation game.

First and foremost, I would like to thank my supervisors, David and Robert, who met my curiosity with enthusiasm and gave me the freedom to explore the topic in my own way, while always providing guidance and support when needed.

I would also like to thank my family and friends for their support in every aspect of life outside the writing of this thesis. The prolonged lunch and coffee breaks and the late nights at Integrerbar are among my fondest memories from my time at the university.

Finally, I must acknowledge the LLMs for their existence, and for being the subject of this thesis. Sadly, they will never understand this thesis, and by the time it is submitted, they will already have forgotten it.

`Alexander`
Sunday 31st May, 2026

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Chapter 1

Introduction

On the 30th of November 2022, OpenAI released ChatGPT [48], a chatbot based on the Generative Pre-trained Transformer (GPT) architecture, which quickly became a viral sensation and sparked a surge of interest in Generative Artificial Intelligence (GenAI) and LLMs. In the following months and years, many other companies released their own GenAI products, such as Google's Bard [54], which later evolved into Gemini, and Anthropic's Claude, making GenAI and LLMs widely accessible to the general public.

With this new technology widely adopted by the general public, it is important to understand how it works, and how it can be used effectively. This raises the question of whether there has been sufficient communication and education about the technology in regards to its capabilities and limitations. Is there a widespread prevalence of misconceptions surrounding LLMs and if so, how can these be addressed?

Bringing together available literature, this thesis proposes a taxonomy of common misconceptions about LLMs, presented in chapter 3. It is divided into four dimensions: Foundational Conceptual Accuracy, Mechanistic Understanding, Ethical and Academic Integrity, and Socio-Cognitive Impact. The taxonomy is intended to be a starting point for further research on the topic.

To address the misconceptions identified in the taxonomy, we have developed a mediation tool in the form of a web application, presented in chapter 4, which provides users with an interactive learning experience. The tool is designed for users with any level of prior technical knowledge, and lets them interact with a simplified LLM to learn about its inner workings and explore how its features influence the output. The features are organized in a skill-tree, where users can choose which features they want to learn about, each presenting a scrollytelling page about the feature.

We performed a small-scale exploratory study to identify misconceptions from the taxonomy in a group of 10 participants through a deductive content analysis of an interview with each participant. We then let the participants interact with the tool, and performed a second interview to identify any changes in the misconceptions. The results of the study are presented in chapter 5, and the implications of the results are discussed in chapter 6.

Chapter 2

Background and Literature

2.1 From Turing to LLMs

'Can machines think?' This question was famously posed by Alan Turing in his 1950 paper "Computing Machinery and Intelligence" [67]. Rather than trying to define what it means for a machine to think, Turing proposed the "Imitation Game", now commonly known as the Turing Test, as a way to determine whether a machine can exhibit intelligent linguistic behavior indistinguishable from that of a human. The game consists of 3 participants: a human interrogator, a human respondent, and a machine respondent. The interrogator is tasked with determining which of the two respondents is the machine, based solely on their responses to questions. The machine, on the other hand, is there to convince the interrogator that it is the human respondent.

As a consequence of this, Turing shifted the discussion from the abstract question of whether machines can think based on their internal cognition, to the more practical question of whether machines can exhibit intelligent behavior through observable linguistic behavior. This emphasis on conversational imitation would many years later be central to both the public perception and the scientific development of AI.

The term "Artificial Intelligence" has its origins in 1955, when John McCarthy quite optimistically proposed a 2 month workshop the following summer, where 10 researchers would gather to solve this "thinking machines" thing [65]. The term was created to distinguish the field from the also up and coming field of cybernetics, which rather than being about developing intelligent systems, is about analyzing how intelligent systems process information. This early optimism is a common theme in the history of AI, with the field experiencing several periods of hype and subsequent disappointment, often referred to as "AI winters" [65]. Importantly, it has always been clear that AI has a tendency to be met with great expectations, which often also deliberately introduce misunderstandings of what AI is and what it can do.

One significant milestone in the public perception of AI was the development of ELIZA by Joseph Weizenbaum in 1964 [23, 65]. ELIZA was a simple program that mimicked a Rogerian psychotherapist, using pattern matching and substitution to create the illusion of understanding. Despite its simple inner workings, ELIZA was able to convince many users that it had genuine understanding and empathy. The phenomenon of users attributing human-like qualities to a machine, is still known as the "ELIZA effect" [65], an effect that is still relevant today.

ELIZA was used as a psychological experiment, in many ways similar to a Turing Test, and for many years AI systems remained primarily confined to research and specialized industrial settings.[1] The public release of ChatGPT at the end of 2022 marked a significant shift in the accessibility and visibility of AI technology. This is clear by looking at Google search trends for the term "AI", shown in Figure 2.1, which shows a significant spike in interest following the release of ChatGPT. This widespread adoption of LLMs introduced new opportunities for interaction with AI systems, but also new challenges in terms of understanding and managing the expectations and perceptions of users.

This raises the question if we have failed in educating lay audiences about the capabilities and limitations of LLMs, and how to best address the misconceptions that may arise from the widespread use of these systems.

In order to address the misconceptions surrounding LLMs, a well-designed mediation strategy is needed. Thus, a didactic model is needed to guide the design of the mediation.

2.2 Didactic Models

The Zone of Proximal Development (ZPD), introduced by Vygotsky [71], describes the space between what a learner can do independently and what they can do with guidance

[1] For a more complete overview of the history of AI, see Strümke [65]. For Natural Language Processing (NLP) specifically, Ghaseminejad [23] provides a comprehensive overview of the evolution from early rule-based systems to modern LLMs.

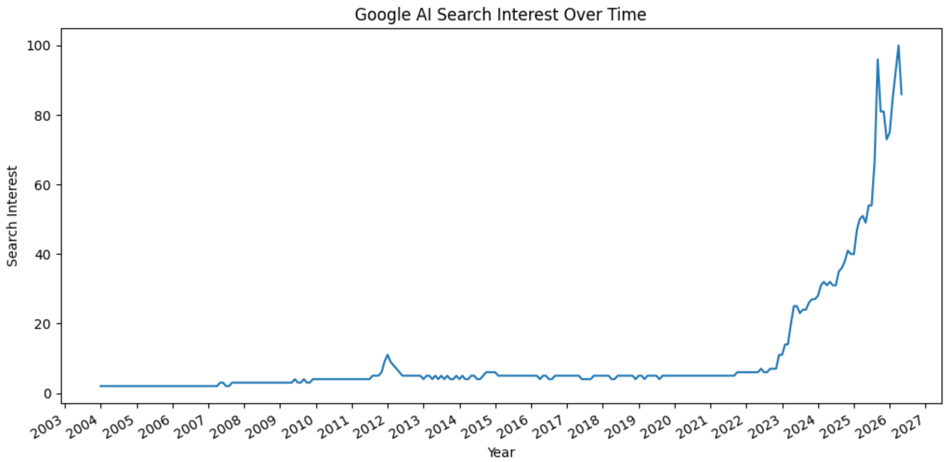
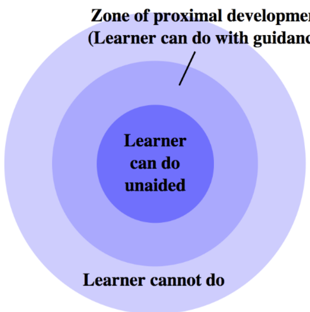
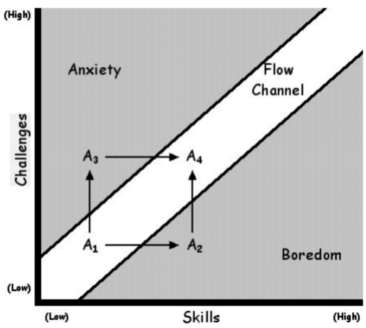


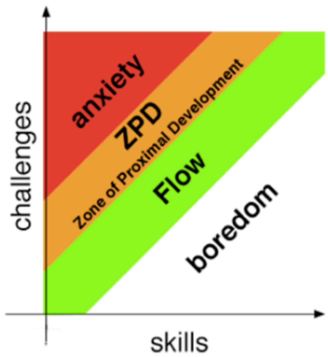
Figure 2.1: Google search trends for the term "AI" from 2004 to 2026. Data retrieved from Google Trends.



(a) The Zone of Proximal Development



(b) The Flow framework



(c) The Zones of Proximal Flow

Figure 2.2: Didactic frameworks. All retrieved from [1]

from a peer or instructor, representing tasks that lie just beyond their current abilities (Figure 2.2a). The framework holds that what is positioned in ZPD for learners today, will be within actual development tomorrow. That is, what knowledge or skill the learner is able to do with some guidance, will eventually be something they can do independently. Therefore, learning is most effective when learning activities are within this zone. As a consequence, the ZPD shifts as competence increases, and the learner is able to perform more complex tasks independently. Similarly, we have the concept of "flow", introduced by Csikszentmihalyi [14], which describes a state of complete immersion and focus in an activity (Figure 2.2b). Flow is accomplished when the challenge of an activity is balanced with the skill level of the individual.

When these two frameworks are combined, we get the concept of the Zones of Proximal Flow (ZPF), shown in Figure 2.2c. ZPF was presented by Basawapatna et al. in 2013 [1], and consists of the flow zone and the ZPD. They argue that learning is most effective when learners are kept within these zones. They back this up with empirical data from classroom studies, which showed that students were more engaged and were enabled to create more complex programs in their game design assignments. The idea is that if the tasks or activities does not challenge the learner enough according to their skill level, they will fall into boredom, and if the tasks or activities are too challenging, they will fall into anxiety. Therefore, it is important to keep learners within the flow zone and the ZPD in order to maximize engagement and learning outcomes.

The framework of ZPF is relevant for the design of the mediation tool, as it provides a theoretical basis for how to structure the learning experience in a way that keeps users engaged and promotes effective learning. For analyzing the misconceptions, we will look at the approach of content analysis.

2.3 Content Analysis

Content analysis is a research method used to analyze and interpret the content of various forms of communication, such as text, audio, or video [19]. It is a technique designed to yield reliable and replicable results [32]. It involves systematically interpreting and analyzing content both qualitatively and, where the research question calls for it, quantitatively, in order to identify patterns, themes, and relationships. This involves coding the content into categories, which can be done either as an inductive or deductive approach. In an inductive approach, categories are derived from the data itself, while in a deductive approach, categories are predefined based on existing theories or frameworks [19]. Involving multiple coders is considered good practice, as it allows inter-coder reliability to be established, which is a measure of the agreement between coders.

In a deductive approach to content analysis, the researcher starts with a predefined set of categories based on existing theories or literature [19]. After preparation, the researcher develops a coding scheme, which is a set of rules for how to assign content to the predefined categories. The researcher then applies this coding scheme to the data, systematically categorizing the content according to the predefined categories. Although categories are defined in advance, there is still some flexibility to the deductive approach as categories may be adjusted or refined as the analysis progresses. A codified scheme is important for ensuring consistency and reliability across multiple coders.

Inter-Rater Reliability

Coding data is used as trusted ground for making inferences, but in order to make content analysis reliable, it is important to ensure that the data is collected with the necessary precautions and a common understanding of what the data represents [32]. When multiple coders are involved in the coding process, it is important to ensure that they apply the coding scheme consistently. This can be assessed through Inter-Rater Reliability (IRR) analysis, which measures the level of agreement between coders. For categorical data, Cohen's κ and Krippendorff's α are common coefficients for measuring IRR. A common measure to improve IRR is to train coders prior to independent coding, for instance by conducting joint coding sessions to develop a shared understanding of the categories.

Cohen's κ

Cohen's κ is a coefficient that measures agreement between two coders who each classify items into mutually exclusive categories [12]. Given a collection of categories, and 2 coders who classify items into these categories, Cohen's κ is calculated as follows:

$$\kappa = \frac{p_o - p_c}{1 - p_c} \tag{2.1}$$

where p_o is the observed agreement among raters relative to the total items, and p_c is the hypothetical probability of agreement by chance.

The value of κ can range from -1 to 1, where a value of 0 indicates the agreement is exactly what would be expected by chance. A positive value indicates agreement, with a value of 1 indicating perfect agreement. Conventionally, values above 0.6 indicate substantial agreement, while values above 0.8 indicate good agreement [49]. The value of 1 is only achieved when both coders classify the same number of items into each category. That requires that the coders find the same number of items in each category, which is not always the case. Therefore, you can still argue that there is agreement even if κ is quite low, by comparing the calculated κ to the maximum possible κ_M, which is calculated as follows:

$$\kappa_M = \frac{p_{oM} - p_c}{1 - p_c} \tag{2.2}$$

Where p_{oM} is the maximum observed agreement, found by summing the minimum of the two coders' probabilities for each category. With this comparison, you can argue that there is some agreement, but the coders have different thresholds for assigning items to categories.

Cohen's κ is criticized for penalizing coders who have agreement on marginal frequencies, as two coders with marginal frequencies must have much higher agreement-rate than two coders with "radically different marginals" [32].

Krippendorff's α

Krippendorff's α is a coefficient that measures the reliability of coding in content analysis [32]. In its most basic form, it is calculated as follows:

$$\alpha = 1 - \frac{D_o}{D_e} \tag{2.3}$$

where D_o is the observed disagreement, and D_e is the expected disagreement by chance. How these are calculated depends on the data. For nominal data, disagreement occurs when coders assign different categories to the same item. For ordinal, interval or ratio data, disagreement is weighted based on the distance between categories. The coefficient normally lands on a value between 0 and 1, where a value of 1 indicates perfect agreement, and a value of 0 indicates agreement and disagreement occur by chance. It can however be negative if the observed disagreement is greater than the expected disagreement by chance, which indicates systematic disagreement [32].

Given 2 coders, a set of categories, and a set of items to be categorized, you can calculate α for each category by using the binary data approach [32] on the presence or absence of each category in the coding. In our study, the items will refer to the participants, and the categories to the taxonomy categories. This process starts by tabulating the encodings for each category with one row for each coder, and one column for each item (shown in Table 2.1). This is called a reliability data matrix. As an example, let us say that we have 2 coders, Alice and Bob, who are coding 8 items on the presence of a category, where 1 indicates that the coder coded the category for that item, and 0 indicates that they did not code the category for that item. Table 2.1 shows how the reliability data matrix would look like for this example.

From these encodings, you can create coincidence matrices for each category as shown in Table 2.2a. Here, o_{00} is the number of items where both coders did not code the category, o_{11} is the number of items where both coders coded the category, and o_{01} and o_{10} are the number of items where only one of the coders coded the category.

Table 2.1: Example of binary coding for 2 coders and 8 items for a single category

	Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8
Alice	1	0	1	1	1	0	0	0
Bob	1	0	0	1	1	0	0	0

For our example, in order to construct the coincidence matrix, we need to take into account all values that the two coders assigned to the items. That is two pairs of values for each item, one for Alice and one for Bob, in total 16 pairs of values for the 8 items. Table 2.1 therefore contains eight 0–0 pairs, so $o_{00} = 8$; one 0–1 pair, so $o_{01} = 1$; one 1–0 pair, so $o_{10} = 1$; and six 1–1 pairs, so $o_{11} = 6$.

Table 2.2: Coincidence matrices in general form and for the example coding in Table 2.1

	0	1	
0	o_{00}	o_{01}	n_0
1	o_{10}	o_{11}	n_1
	n_0	n_1	n

(a) General form

	0	1	
0	8	1	9
1	1	6	7
	9	7	16

(b) Example

For binary nominal data with two coders, disagreement occurs when one coder assigns 0 and the other assigns 1. The observed disagreement is therefore proportional to the number of off-diagonal entries in the coincidence matrix.

Using the binary coincidence formulation presented by Krippendorff [32], the coefficient can be simplified to:

$$\alpha = 1 - (n-1)\frac{o_{01}}{n_0 n_1} \tag{2.4}$$

In the example with Alice and Bob, we have $n = 16$, $n_0 = 9$, and $n_1 = 7$. Substituting these values into the formula gives:

$$\alpha = 1 - (16-1)\frac{1}{9 \cdot 7} = 1 - \frac{15}{63} \approx 0.762 \tag{2.5}$$

In this project, the `krippendorff` library for Python [8] was used to calculate α. While the library implements a generalized, vectorized algorithm to optimize computational efficiency, it represents a mathematically equivalent formulation to the manual method described above.

With this background in place, as well as a didactic framework and a method for analyzing the misconceptions, we can now look at the misconceptions themselves, and how they can be categorized in a taxonomy.

Chapter 3

Misconceptions Taxonomy

The field of studying current LLM misconceptions is vast and rapidly evolving, but is still in its early stages, and there is no widely accepted taxonomy of misconceptions. In order to systematically analyze the misconceptions surrounding LLMs, and the effects of the developed mediation tool, we have developed a taxonomy of misconceptions. This taxonomy is based on a review of the literature on LLM misconceptions, as well as an analysis of our own experiences and observations.

Taxonomies are classification systems aimed at helping to organize and understand complex phenomena [33]. Their use is wide spread to provide a framework for classification, identification and description of phenomena [62].

Existing research and literature on misconceptions surrounding LLMs has identified several key dimensions. We have chosen to divide the taxonomy into four dimensions, each of which captures a different aspect of LLM misconceptions. These dimensions are: foundational conceptual accuracy, mechanistic understanding, ethical and academic integrity, and socio-cognitive impact. Each dimension is further divided into subcategories, which capture specific types of misconceptions within that dimension. Let us first take a look at misconceptions related to foundational conceptual accuracy, which is the first dimension of the taxonomy.

Table 3.1: Combined Misconceptions Taxonomy by Dimension

Dimension	Label	Description
Foundational Conceptual Accuracy	ANTHRO	Attributing human mental states, intentions, or understanding to LLMs.
	DETAUTO	Confusing LLMs with deterministic automation systems.
	OVERACC	Overestimating the reliability and correctness of LLM outputs.
	OVERAUTO	Belief that the LLM operates independently or possesses agency.
Mechanistic Understanding	REALTIME	Belief that LLMs retrieve current (real-time) information.
	HALLBLIND	Failure to recognize that LLMs can produce false or fabricated information.
	CONTEXNEG	Failure to understand the limitations of the context window.
	SEMUNDER	Lack of awareness that the LLM processes language as tokens rather than semantic units.
	SOURCEATTR	Belief that LLM output originates from specific identifiable sources.
	CONDTREAT	Belief that LLMs process prompts differently based on content or context.
Ethical & Academic Integrity	PLAGBLIND	Failure to recognize ethical/legal issues related to traceability of sources behind LLM-generated content.
	COPYBLIND	Failure to recognize that LLM-generated content may be subject to copyright/usage restrictions.
	CONDACC	Ethical acceptance of LLM use only under certain conditions or domains.
	ENVNEGL	Failure to recognize the environmental impact of LLM training and operation.
Socio-Cognitive Impact	REPLFEAR	Belief that LLMs will replace human roles, expertise, or professions.
	COGPASS	Tendency to rely on LLMs rather than engaging in independent reasoning or critical thinking.
	DEPRISK	Failure to recognize long-term reliance risks, skill degradation, or loss of autonomy.
	CREDBIAS	Treating LLM outputs as inherently truthful or authoritative without critical evaluation.

3.1 Foundational Conceptual Accuracy

This dimension of the taxonomy is focused on misconceptions related to the foundational conceptual understanding of LLMs, and also the confusion between LLMs and other types of AI systems. The term “AI” exploded in popularity after the release of the GPT-3.5 model[1], and the term is often used to refer to a wide range of technologies, including LLMs. A 2023 study by Santos et al. [59] showed that people were familiar with the term “AI”, but less familiar with associated terms such as “neural networks”. This has lead to a lot of confusion and misconceptions about what LLMs are and what they can do.

3.1.1 Anthropomorphism – ANTHRO

Anthropomorphism is defined as the attribution of human mental states, intentions, or understanding to LLMs. This misconception seems to be common amongst users who interact with LLMs in a conversational manner, like chatbots. Subjects was placed in this category by using terms such as “understand”, “think”, “learn” and “know” to describe the LLM’s capabilities in a non-metaphorical way.

Anthropomorphism emerges both from the system design of LLMs, and from the user’s interpretation of the system’s behavior. A paper by Maeda and Quan-Haase [41] provides a framework for when Human-AI Interaction (HAI) become parasocial. They argue that chatbots’ variety in information presentation resemble day-to-day bi-directional communication, and that the users’ tendency to fill in missing conversational context causes them to attribute human-like qualities to the system. The roleplaying nature of the chatbot interface, which is visible through language choices such as personal pronouns and paraphrasing, suggests attributes such as intention and understanding in the system which lead users to project inference out to the system due to the lack of information about the system’s inner workings.

This is empirically supported by a large scale study by Ibrahim et al. [28] which found similar human-mimicking behaviors in all tested LLMs (Gemini 1.5 Pro, Claude 3.5 Sonnet, GPT-4o and Mistral Large). They found, in particular, use of personal pronouns, validation and empathy as the most common anthropomorphic behaviors, and that these behaviors mostly occurred after multiple turns of conversation.

[1] As shown in Figure 2.1

The anthropomorphism misconception is widespread. An investigation by Colombatto and Fleming [13] in 2024 found that in a sample of 300 US residents, only a third of the participants thought that ChatGPT definitely did not have subjective experiences, while two thirds applied varying degrees of phenomenal consciousness. The investigation also showed that the participants' consciousness attributions increased with usage frequency, rather than higher use leading to a corrected misconception.

Building on this, Lin [38] conceptualizes anthropomorphism as an "anthropomorphism fallacy", in which users and researchers misinterpret LLM outputs as evidence of underlying mental life. The fluent, human-seeming responses of LLMs can lead to an enhanced Eliza effect, encouraging users to attribute understanding, beliefs, or intentions to systems that in reality operate through statistical token prediction. This creates a "language-user illusion", where the system's linguistic capabilities are mistaken for genuine cognition.

Furthermore, the anthropomorphic metaphors in use surrounding LLMs can reinforce this misconception [38]. Unlike humans who are influenced by their own experiences, LLMs simulate a wide range of viewpoints based on their training data [38], which can lead to overestimation of their capabilities.

3.1.2 Deterministic Automation – DETAUTO

Deterministic automation is defined as the confusion between LLMs and deterministic automation systems. One could say that this misconception is placed on the opposite side of the spectrum compared to the previous misconception of anthropomorphism. Before the widespread use of LLMs, one could often find deterministic automation systems in customer service chatbots, which were designed to follow a fixed set of rules and provide predetermined responses. It seems natural that users would initially confuse LLMs with these deterministic systems, especially since they are often presented in similar interfaces. Subjects were placed in this category if they described LLMs as following a fixed set of rules, or if they expressed surprise at the variability and creativity of LLM outputs. The expression of a belief that the LLM contained a database was not categorized here, as this misconception covers the model's process rather than its memory design.

This misconception is likely reasoned by a lack of interaction experience with LLMs. In a study performed by Schneider in 2025, analyzing over 200 000 conversations, he found that users initially approached LLMs as traditional software tools [61]. Initial prompts were often structured and machine-like, but already on the second turn, users shifted towards more natural, conversational interaction. This was shown by the use of a less complex language, use of politeness indicators, shorter prompts and more personal interaction.

A consequence of this misconception arises when LLMs actually replace deterministic automation systems, leading to failures in safety-critical deployments. Vinay [70] argues that LLM failure patterns "differ fundamentally from those of traditional machine learning models" where deterministic inference processes are assumed. A lack of awareness of these differences can lead to failures going undetected. For developers, automatic code generation tools produce syntax violations, invalid reference errors and Application Programming Interface (API) knowledge conflicts in high rates, which are failures deterministic systems do not produce [51].

This misconception could also work as a springboard to other misconceptions covered in this taxonomy. Patterns of logical reasoning could lead to an overtrust in LLMs leading to placements in categories of OVERACC and CREDBIAS [21, 43].

3.1.3 Accuracy Overestimation – OVERACC

Accuracy overestimation is defined as the overestimation of the reliability and correctness of LLM outputs. This misconception is likely correlated with other misconceptions in this taxonomy, where failure to understand fundamental concepts and mechanics of LLMs can lead to an overestimation of their capabilities. Subjects were placed in this category if they expressed a belief that LLMs are more accurate or reliable than their design justifies, or if subjects failed to recognize the potential for errors and inaccuracies in LLM outputs.

The most fundamental reason for this misconception is automation bias, which is the tendency for people to trust in automated systems even when they contradict available evidence. In this taxonomy, automation bias is covered in CREDBIAS. Carnat[7] argues that this phenomena spawns from ANTHRO. The misconception is widespread, shown by a experiment by Klingbeil et al. [31] in 2024, where they found that the mere knowledge of advice being generated by an AI causes people to over-rely on it. This is the case even when the advice contradicts available contextual information and their own assessments.

Accuracy overestimation is even present in the LLMs themselves. In 2025, Sun et al. found that after testing the confidence calibration of five different LLMs, they overestimate their accuracy by 20% to 60% [66]. Naderi et al. also found similar results saying that LLMs retained high confidence regardless of the correctness of their outputs [42].

A rather obvious consequence of this misconception is the spreading of misinformation. The convincingly phrased output of LLMs are often perceived to be valid, trustworthy and satisfactory, and users also show a high tendency to follow LLM advice provided [63].

3.1.4 Autonomy Overestimation – OVERAUTO

The category of autonomy overestimation is defined as the belief that the LLM operates independently or possesses agency. This category was was inspired by a study performed by Wang et al. in 2025, where they researched the mental models of GenAI chatbot ecosystems [72]. In this study, participants were asked to draw a diagram of how they thought the chatbot ecosystem worked when they used it to book a hotel room. The study showed varied mental models of the degree of autonomy of the GenAI in the ecosystem. Participants placed in this category expressed beliefs that the LLM operates independently, or "acts" on behalf of the user.

It should be said that there does exist agentic tools which utilizes LLMs as a component. For instance, the integrated Copilot in Microsoft Visual Studio Code is an agentic tool, as the user can allow shell actions where the LLM is used to generate commands based on the user's prompt, and then executes those commands in the terminal without further user interaction or confirmation. This gives the user the experience of the LLM browsing the files in their project, writing code and executing it, testing the code and debugging it. This can be experienced by the user as if the LLM possesses full autonomy, but it is actually the surrounding infrastructure that is making this possible, not the LLM itself. It is a misconception to not make this distinction.

It is likely that this misconception spawns from the ANTHRO misconception, as it is natural to assume a real social actor when presented with a system that produces natural language outputs[55, 30]. Thus, there is expected to be a high level of correlation between these two categories.

3.2 Mechanistic Understanding

The second dimension of the taxonomy is focused on misconceptions related to LLM-specific mechanics. Most modern LLMs today are transformer-based models [69], and not being aware of the internals of how these models work can lead to misconceptions about their capabilities and limitations. This dimension differs from the first dimension of foundational conceptual accuracy, as it is focused on the specific mechanics of LLMs, rather than misconceptions about LLMs on a higher level of abstraction.

3.2.1 Real-Time Data Belief – REALTIME

LLMs are trained on large, fixed datasets, and generate responses based on patterns learned from that data. Therefore, mistakenly believing that LLMs can access real-time information or the internet is a common misconception. People placed in this category expressed beliefs that LLMs will gather information from the ongoing conversation and that will "update" the model's knowledge outside of what can be placed in the context window. This could be expressed as the belief that the model will learn from the conversation[2] currently had with the user and will be able to utilize that information in future conversations, or that the model can learn something from another conversation with another user and utilize that information in a conversation with the first user.

Live access to the internet is also under the category of real-time data belief, as LLMs by themselves do not have the capability to access the internet or retrieve real-time information. The introduction of Retrieval-Augmented Generation (RAG) [35] has somewhat blurred the lines here, as it allows LLMs to retrieve information from external sources, but it is still a misconception to believe that LLMs have this capability by default. It is still necessary to be aware of the limitations of RAG systems, as the underlying LLM's response generation is based on the retrieved information being placed in the context window. [73].

3.2.2 Hallucination Blindness – HALLBLIND

The term "hallucination" in the context of LLMs refers to a situation where the output generated by the model is deemed as factually incorrect. Many believe that this is an exceptional state of the LLM, rather than the very functionality of it. The correctness of a response is subject to the reader's interpretation and not a feature of the LLM itself. Failure to make this distinction is what we have categorized as hallucination blindness, which is defined as the failure to recognize that LLMs can produce false or fabricated information.

The theoretical backing for this claim is well established in the literature. Bender and Koller [3] argue that LLMs, which are solely trained on "linguistic form", have no mechanism to acquire "meaning", where meaning is defined as the relationship between linguistic form and the communicative intent. Because the training data used for LLMs

[2] In a chatbot environment

only consists of text, the model has no way of learning the non-linguistic reality those texts refer to, but only the patterns in the text itself. The model only learn the "projection of communicative intent." This is extended by Bender et al. [4], which characterizes LLMs as "stochastic parrots", stitching together sequences of linguistic form according to probabilistic patterns. The meaning and coherence of the generated output resides not in the model itself, but rather the receiver of the information.

With this in mind, hallucination is no malfunctioning at all, but rather exactly what the model is designed to do. Having a form of "correctness" requires a correspondence between language and the outside world, which the model does not have and therefore cannot evaluate their own output [3]. Hallucination is inevitable [23]. Failure in realizing that hallucination is, and will continue to be, a feature of LLMs may cause users to also be categorized into other misconceptions such as OVERACC or CREDBIAS.

There is an argument to be made that the use of RAG will mitigate generation of incorrect information by grounding the outputs in retrieved information, which is a meaningful improvement to the technology. However, it does not solve the underlying issue as a RAG-model still generates its final output through the same token prediction mechanism. It can still misrepresent, misquote, or fail to utilize the retrieved information and can in no way verify the truthfulness of what it generates. Hallucination remains possible whenever the generation process outruns the retrieval.

3.2.3 Context Window Neglect – CONTEXNEG

LLMs operate within a fixed sized context window, which limits the amount of information they can process at once. That is, you can only input a certain amount of tokens into the model. Subject who showed no indication of understanding this limitation, or who expressed beliefs that LLMs can process unlimited amounts of information, were categorized into context window neglect.

Even within the context window, there are limitations as to how much information the model can effectively utilize. According to a study performed by Liu et al. in 2023, LLMs shows much higher performance if the relevant information is placed at the beginning or the end of the input context [39]. Even if all the relevant information is retrieved correctly from the context, LLMs performance still degrades noticeably as the amount of information increases [18]. A study done by Paulsen in 2025 showed that given input of datasets of varying sizes, the top-line LLMs failed to provide correct answers when the context dataset size exceeded 100 tokens, and that the accuracy plummeted when the context dataset size exceeded 1000 tokens [50].

Failure to understand the limitations of the context window could be strongly correlated with other misconceptions in this taxonomy, such as OVERACC, REALTIME and CREDBIAS, as it is likely that users who do not understand the limitations of the context window will also overestimate the accuracy of LLM outputs, and believe that LLMs can access real-time information. This corelation seems to common amongst students using LLMs for academic purposes [46, 57].

3.2.4 Semantic Understanding – SEMUNDER

LLMs process language as tokens rather than semantic units. This means that the model does not understand language in the way humans do, but rather as a sequence of many-dimensional vectors, whose values are used to calculate the probability of the next token in the sequence. This is done through first splitting the input text into tokens, which can be done on different levels, such as words, subwords or even characters. Most modern LLMs use subword tokenization as this has shown a higher level of performance while still being able to handle tokens not in their vocabulary.

After the tokenization process, the tokens are converted into embeddings through a pre-trained word embedding model. A word embedding model is a machine learning model that is trained to represent tokens as vectors in a high-dimensional space, where semantically similar tokens are represented by similar vectors. The idea is that the model can capture relationships between tokens based on their co-occurrence in the training data. This purely done through statistical patterns in the training data, and does not involve any information of the meaning of the tokens.

Understanding text semantically is a human ability which cannot be fully captured by a statistical model [3, 2]. Therefore, it is a misconception to believe that LLMs understand language in the same way humans do. Subjects who expressed beliefs that LLMs have a semantic understanding of language, or who failed to recognize that LLMs process language as tokens, were categorized into SEMUNDER. There should be a high level of corelation between this misconception and the ANTHRO misconception, as it is likely that users who anthropomorphize LLMs will also believe that they have a semantic understanding of language. Since the output of LLMs also are embeddings turned back into tokens, it is likely that users who do not understand the tokenization process will also fail to understand that the output is also generated based on token patterns rather than semantic understanding, which could lead to placements in the OVERACC and CREDBIAS categories.

3.2.5 Source Attribution Error – SOURCEATTR

Since LLMs are trained to mimic the patterns in their human-written training data, it comes as no surprise that the generated output will resemble how humans write and cite their sources. The generated output of LLMs is still limited to the statistical patterns in the training data, and even if the output resembles a citation, it does not mean that the LLM is actually retrieving information from specific identifiable sources. For RAG systems, the retrieved information is still processed based on token patterns, and the LLM does not acquire an understanding of the information it retrieves. One could argue that the use of RAG systems does make the citation identifiable and thus the information is retrieved from specific identifiable sources, but the information is still processed based on token patterns.

Once the LLMs have gone through training, it is also impossible to trace back where generated information comes from in the training data, as the model weights do not store information in a way that allows for such tracing. Subjects who in some way expressed belief that LLM output originate from specific identifiable sources, or to some degree expressed that the information is learned in training, were categorized into source attribution error.

The mimicking of citation patterns by LLMs can lead to an overestimation of the reliability of the generated information and therefore lead to placements in the OVERACC category. A study by Ding et al. in 2025 found that users deemed responses with citations as more reliable than those without, even if the citations were randomly generated. Fact checking these citations led to a significant decrease in trust in the response [16]. This result is backed by Do et al. who found that output with citations were deemed more trustworthy than output without [17].

3.2.6 Conditional Treatment – CONDTREAT

Every prompt given to a LLM is processed based on the same underlying mechanics, which are the statistical patterns in the training data. The length, or the semantic complexity of the prompt does not change the underlying mechanics and therefore it is a misconception to believe that LLMs process prompts differently based on the content or context. Subjects who in some way expressed belief that LLMs will process one prompt differently than another were placed in this category.

It is possible that modern Chain of Thought (CoT) techniques can contribute to this misconception. CoT is a prompting technique where the model is prompted to generate a step by step evaluation of the original prompt. LLMs in chatbot environment often hide the CoT process from the user, showing a buffering text like "thinking about the problem" or "finding the best solution" before displaying the final part of the output. It is easy to see how this could lead to the belief that the model is processing the prompt differently based on the content or context, as it gives the impression of a more complex and dynamic processing of the prompt. This is however, still the same token prediction mechanism.

3.3 Ethical and Academic Integrity

The third dimension of the taxonomy shifts focus away from the what and how of LLMs, and instead focuses on issues related to the ethical and academic integrity of LLM use. The use of LLMs in academic settings has raised a lot of ethical questions, and there is an ongoing debate on whether LLMs have a place in academia at all, and if so, under what conditions. There are also ethical questions related to the use of LLMs outside of academia, such as the environmental impact of training and operating LLMs.

3.3.1 Plagiarism Blindness – PLAGBLIND

A core idea in academia is that the ancestry of ideas and information should be traceable to their sources, and that proper attribution should be given. This is important for maintaining academic integrity and giving credit where it is due. However, since there is no way to trace exactly where the generated information comes from in the training data, it is a misconception to believe that LLM-generated content is free from ethical and legal issues related to plagiarism. If a subjects show little to no awareness of the ethical and legal issues related to the traceability of sources behind LLM-generated content, they should be categorized into plagiarism blindness.

A common use case of LLMs is to generate ideas for starting a project. At the Faculty of Social Sciences at The University of Bergen (UiB) guidelines have been adopted for the use of LLMs in academic work [68]. These guidelines are divided into five categories, where category 1 is the most strict with no use of LLMs allowed, and category 5 is the most lenient with no restrictions on the use of LLMs. The second most strict category, category 2, states that LLMs can be used for idea generation and information search. This seems troublesome in regards to plagiarism, as the generated ideas and information cannot be traced back to their sources in the training data. The fact that the faculty has adopted these guidelines, could contribute the the lack of awareness surrounding these ethical questions. Naithani [44] suggest that both responses and ideas generated by a LLM should be attributed to the LLM as the author, but also raises the question of whether LLMs has the moral right to attribution.

3.3.2 Copyright Blindness – COPYBLIND

This misconception differs from plagiarism blindness, as it is focused on the ownership of the generated content, rather than the traceability of sources. Multiple jurisdictions have already concluded that GenAI output does not meet the requirements for copyright protection, as copyrightability requires human creative involvement [20, 40]. This raises the question of who actually owns rights in the generated content. The LLM itself, the user who provided the prompt, the developer of the LLM or the author of the training data? If a subject fail to discuss the ownership of LLM-generated content, or express beliefs that LLM-generated content is not subject to copyright and usage restrictions, they should be categorized into copyright blindness.

3.3.3 Conditional Acceptance – CONDACC

The category of conditional treatment, where users express beliefs that LLMs process prompts differently based on the content or context, can develop to the category of conditional acceptance. Some users may express that they are comfortable using LLMs for certain tasks, such as idea generation or information search, but not for others, such as writing a paper or generating code. In the academic setting, this can also be expressed as acceptance of LLM use only in certain domains. This could be for instance claiming that LLMs are better at programming rather than physics. This is possibly backed by data from a Statista global survey, stating that only 16% expressed belief that AI adoption would impact their job for the worse, while twice as many (36%) expressed belief that AI adoption would negatively impact the job market [10]. Subjects who expressed beliefs that they are comfortable using LLMs for certain tasks but not for others, or who expressed acceptance of LLM use only in certain domains, were categorized into conditional acceptance.

3.3.4 Environmental Neglect – ENVNEGL

The environmental impact of AI and specifically LLMs is already widely reported in mainstream media [29, 34, 45, 60]. We see it as a misconception to fail to recognize the environmental impact of LLM training and operation. If subjects show little to no awareness of the environmental impact of LLM training and operation, they should be categorized into environmental neglect.

The scale of the energy consumption of LLMs is huge. Training LLMs already requires a significant amount of energy, the GPT-3 model, for instance, was estimated to consume 1,287 MWh of energy during training alone [15]. However, the energy consumption of LLMs does not stop after training, as about half of the energy consumption of LLMs comes from inference, which is the process of generating responses to user prompts [15, 47]. Beyond just energy consumption, the water footprint of AI is also significant. Just the training of the GPT-3 model in Microsoft's United States data centers was estimated to directly evaporate 700 000 liters of water for cooling purposes [36]. At inference, the same model consumes half a liter of water for every 10–50 medium length responses [36].

A logical reference point would be to compare LLM-prompting with an ordinary web search. In early 2023, Alphabet's chairman suggested that interacting with a LLM is likely to cost ten times more energy than a web search [15]. Given that a standard Google search is estimated to consume around 0.3 Wh of energy [25], that would mean that a single LLM prompt could consume around 3 Wh of energy. This is consistent with the per-request estimated energy consumption of ChatGPT [15]. Scaling this up to the number of daily users, the annual energy consumption of Google's AI alone is comparable to the annual energy consumption of a country like Ireland [15].

3.4 Socio-Cognitive Impact

The fourth and final dimension of the taxonomy is focused on misconceptions related to the socio-cognitive impact of LLMs. The widespread use of LLMs has raised concerns about their potential impact on human cognition, and society as a whole. Some of these concerns are warranted, while others are limited to misconceptions about how LLMs work and their capabilities.

3.4.1 Replacement Fear – REPLFEAR

It remains to be seen how LLMs will impact the job market, but there is belief amongst some that LLMs will replace human roles, expertise, or professions. Multiple surveys have shown that about a third of the population express fear of job replacement by AI [10, 37]. However, more then half of US workers found it unlikely that their job would be replaced by AI, but rather change how they do their current job [37].

Where this fear becomes unwarranted, and where a misconception arises, is when people shifts from "LLMs can replace certain tasks" to "LLMs can replace human roles". Subjects who expressed this belief was placed in the category of replacement fear.

3.4.2 Cognitive Passivity – COGPASS

This misconception is defined as the tendency to rely on LLMs rather than engaging in independent reasoning or critical thinking. This misconception was not categorized based on the subjects' beliefs about the capabilities of LLMs, but rather based on their responses on how they themselves describe their own use of LLMs.

That raises the question of what sort of habits and behaviors that would be categorized into cognitive passivity. The habits most likely to create cognitive passivity are the ones that make the LLM a first-resort for information and problem-solving, and as a substitute for thinking about the problem at hand. That is, if a user describes that when they encounter a problem, they start of by asking a LLM for help, or if they describe immediate use of LLMs when comparing different approaches.

A study performed by Gerlich in 2024 found that there is a high level of negative correlation between frequent use of AI-tools and critical thinking skills [22]. The mediator of this correlation was found as the increase in cognitive offloading. When users were introduced to AI-tools might be critical, as the study showed that younger participants were more likely to offload cognitive tasks to the AI-tool, which resulted in lower scores in critical thinking. However, the study also showed that a higher level of education was a protective factor against this effect, as higher educated participants were more likely to cross-check and evaluate responses from the AI-tool.

3.4.3 Dependency Risk – DEPRISK

While the previous misconception of cognitive passivity is assigned based on the subjects' descriptions of their own use of LLMs, the category of dependency risk is assigned based on the subjects' ability to discuss the potential long-term consequences of extensive LLM use. Subjects who, in describing their use of LLMs, showed no to little recognition of the risk of becoming reliant on LLMs, or described tasks that should not require LLM use, but still used them for those tasks, were categorized into dependency risk.

As discussed in the previous category, there is a negative corelation between frequent use of AI-tools and critical thinking skills [22], and this becomes more coherent when it gets to the point of cognitive offloading. Cognitive offloading is the act of shifting tasks with cognitive demands onto external tools. There has been shown an impairment between cognitive offloading and confidence calibration, while the most accurate confidence calibration is found amongst unaided decision makers [9]. This springs into other potential consequences of dependency risk, such as spread of misinformation as a result of OVERACC.

3.4.4 Credibility Bias – CREDBIAS

This misconception is based on older automation bias research, which shows a over-reliance on automated systems, or a higher trust in automated systems [24]. If LLMs are used as decision support systems, there is a risk that this automation bias will be transferred to LLMs, leading to a credibility bias, which is the tendency to treat LLM outputs as inherently truthful or authoritative without critical evaluation. Subjects wo in someway that they treat LLM outputs as generally more truthful or authoritative than other human sources of information, or who showed no to little critical evaluation of LLM outputs, were categorized into credibility bias. Being bias towards LLM outputs should be closely aligned to overestimation of LLM accuracy, and therefore there could be a high level of correlation between this misconception and the OVERACC misconception.

With the taxonomy established, chapter 4 covers the methodology of how we approached in developing the mediation tool and analyzing the prevalence and shifts of these misconceptions.

Chapter 4

Methodology

4.1 The Mediation Tool

In order to address potential misunderstandings about LLM-tools, we developed an interactive mediation tool aimed at teaching users the inner workings of LLMs. The intention was that by providing users with a viable mental model of how LLMs work, they would be better equipped to use them in a responsible manner. We wanted to create an interactive tool that would engage users in a way that would be effective in showing them how the different components and aspects of LLMs work together to produce the output they do. At the same time, the tool should be accessible to a wide range of users, not relying on previous technical or theoretical knowledge in Computer Science (CS) or AI.

The intention was that participants in the study would use the tool to learn about LLMs in their own time and at their own pace, without the need for instruction or guidance from us. Thus, it needed to be intuitive and engaging enough to encourage users to explore and learn on their own, keeping them inside the Zones of Proximal Flow (ZPF) [1]. We settled on a web-based tool containing a "build your own LLM"-type experience, where users would be able to themselves choose and configure different components of a LLM and see how their choices would affect the output of the model. The technical implementation of the tool was done as a part of the Informatics Project II course at UiB, independent of this study, with David Grellscheid as supervisor. The tool is currently running on `https://llm.grelli.org/`.

4.1.1 Features

The tool consists of various features and views designed to keep the users engaged and in flow [14]. Each component build upon the users current knowledge to put them in the Zone of Proximal Development (ZPD) [71]. Starting with the most important feature, we have the simple dynamic language model.

The Dynamic Model

The tool consists of various ways to interact with a very simple representation of a LLM. This model is modular, letting the users themselves choose which components and features they want to add to their model configuration. The model itself is based a simple mapping between a token or sequence of tokens and the probability distribution of the next token. There are different versions of this mapping, depending on which features the user has selected in their configuration. There are 3 different tokenizers, 3 different context window sizes and a free choice of up to 2 out of 9 different books from Project Gutenberg [27]. The model also contains a switch for whether the model should be deterministic by selecting the most probable token as the output, or non-deterministic by sampling from the probability distribution.

Build Your Own LLM

The core feature of the tool is the "Build Your Own LLM"-experience. In this view, users will be able to interact with a Graphical User Interface (GUI) containing a prompt-bar and a skill-tree of different components and configurations that they can choose to add to their model configuration. The prompt-bar will allow users to input prompts and then see the output of their model configuration in real-time, showing them the prediction for the next token as ghost text. The skill-tree displays all selected features in the current configuration, as well as all available features that can be added to the configuration. The features are organized in a tree structure, in order to demonstrate the dependencies between different components and configurations. For example, you cannot select the 3-context-window feature without first selecting the 2-context-window feature.

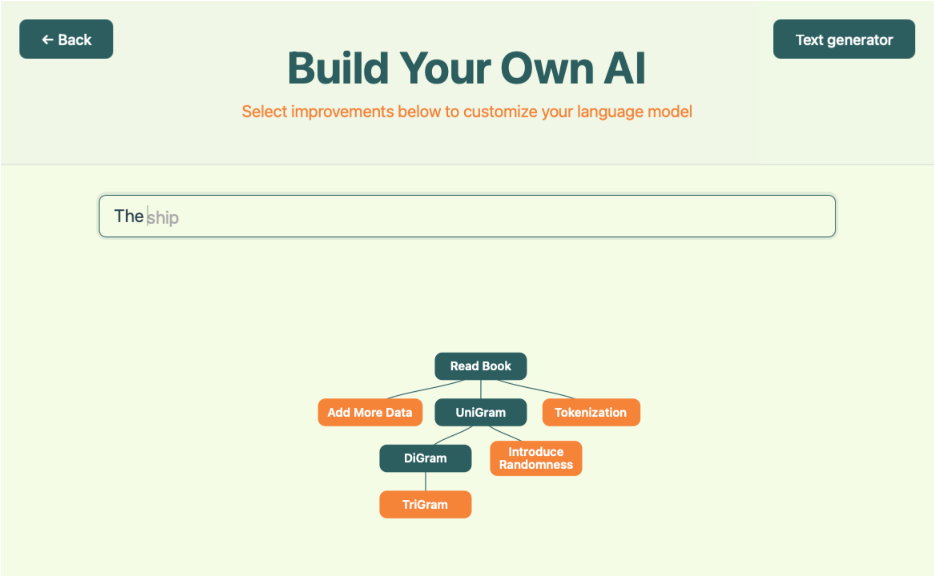


Figure 4.1: The “Build Your Own AI”-view of the mediation tool.

Text Generator

At any point in the “Build Your Own LLM”-experience, users can explore their current configuration in a text generation context. By clicking the “Text Generator”-button, users are taken to a different view containing a larger text input field and a button saying “Generate Text”. Here, users can input longer prompts and at any point click the “Generate Text”-button to have their current model configuration continuously generate the next token in the text, like a normal LLM would. This also demonstrates to the users that the same underlying process can be used for multiple purposes, and that the different features they have selected in the skill-tree work together to produce the output of the model.

Scrollytelling Chapters

Connected to each feature in the skill-tree is a scrollytelling chapter that explains the feature in more detail. Scrollytelling is a form of interactive storytelling where the content visible on the screen changes as the user scrolls down the page. This is utilized in the mediation tool to provide users with a more engaging and interactive way to learn about the different features and components of LLMs. Each chapter consists of a collection of text-boxes, to each of which have a corresponding visual animation that is triggered when the textbox is scrolled into view. The animations are designed to visually demonstrate the concepts explained in the text-boxes, providing users with a more intuitive understanding

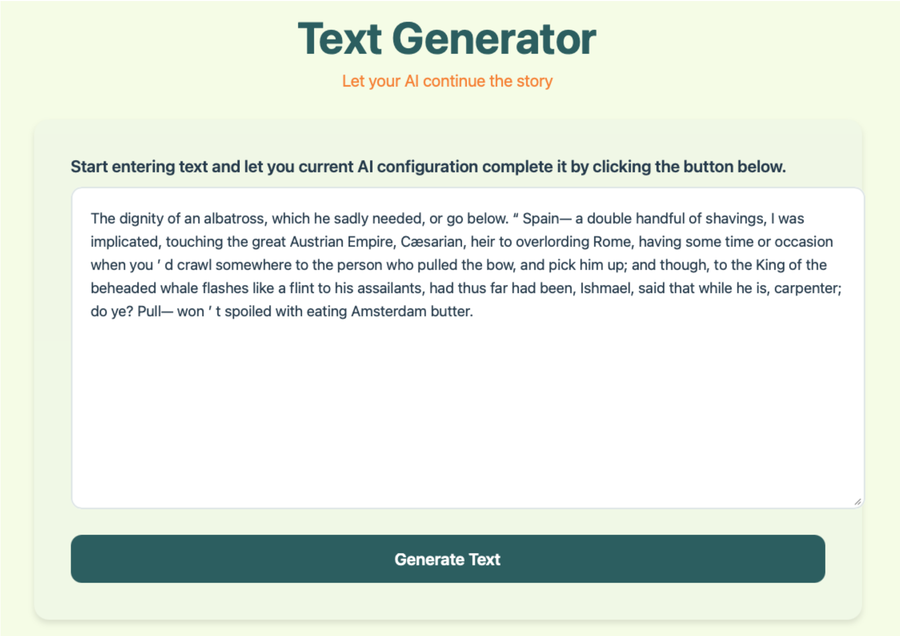


Figure 4.2: The "Text Generator"-view of the mediation tool.

of how the different features and components of LLMs work together to produce the output they do.

Chapter Overview

Users can at any point access each of the individual scrollytelling chapters through a chapter overview page. This page contains a list of all the features available in the tool, organized by category. Each feature has a short description, and by clicking on the feature, users are taken to the corresponding scrollytelling chapter where they can learn more about the feature and how it works.

External Resources

At the bottom of the chapter overview page, there is a section containing a list of external resources for users who want to learn more about LLMs and how they work. This includes online courses, educational videos, articles and blog posts on relevant topics [5, 11, 26, 53, 58, 60, 64].

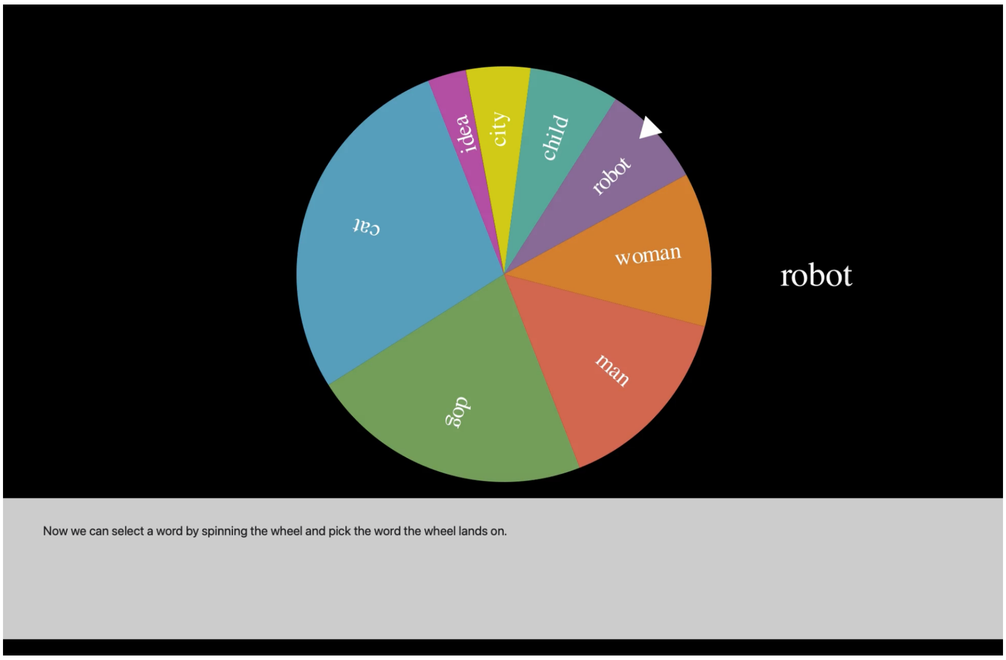


Figure 4.3: The scrollytelling chapter connected to the weighted random sampling feature.

4.1.2 LLM Representation

The mediation tool does not actually contain a real transformer-based LLM under the hood, but rather a simple representation of one. Each branch of the skill tree corresponds to a different feature or component of LLMs. Most of the topics contain multiple chapters building on each other, to keep users in the ZPF [1]. The features are grouped into 4 different categories: training data, context, non-determinism and tokenization. Each of the categories are aimed at giving intuition on how the components work rather than being 100% technically accurate, and the different features are designed to demonstrate the effect of that component on the output of the model. The selection tree, with corresponding topic, can be seen in Figure 4.4. The components selected for mediation are mainly chosen to address misconceptions in the dimension of Mechanistic Understanding, in hope that by giving users a better understanding of how LLMs work, they will be better equipped to use them in a responsible manner.

Training Data

This category is perhaps the most true to real LLMs, as the features in this category actually modify the training data of the model. By understanding training data, you get an understanding on how LLMs are fixed in their setup and do not gain new knowledge

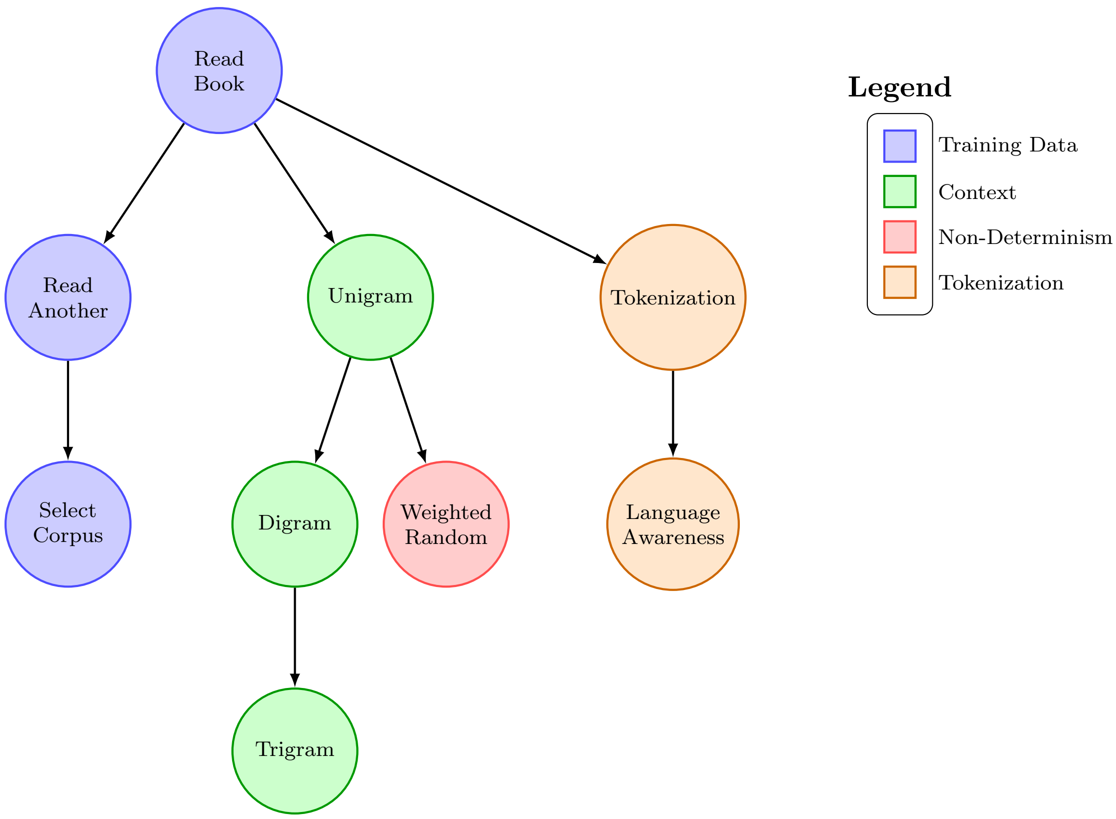


Figure 4.4: Selection tree with grouped LLM features

with interaction from the user. Thus, it will address the REALTIME misconception. Also, by understanding that the tokens used for training have no tracing to where in the training data they come from, it will address the SOURCEATTR misconception.

The first feature the users can add is the "Read Book"-feature, which simply adds a book from Project Gutenberg [27] to the training data of the model. Before adding this feature, the model works by randomly selecting a a word from a 10 000-word english vocabulary as the next token. After adding the feature, the model will only select words from the book that was added to the training data. The learning target of this feature is to introduce the idea that models are trained on previously written human text, in the hope that it will learn the patterns and structures of human language.

Secondly, users can add the "Read Another"-feature, which adds a second book to the training data. This will concatenate the two books together, and the model will select the next token from either of the two books. The learning target of this feature is to show that to get LLMs to generalize, they need to be trained on a large and diverse dataset, containing many different examples of human language.

Finally, users get the option to select which books they want to add to the training data with the "Select Corpus"-feature. This feature does not alter the underlying process of the model, but it allows users to explore how different training data can affect the output of the model. All books available in the tool are from Project Gutenberg [27].

Context Window

The context window of a LLM is the number of previous tokens that are taken into account when predicting the next token. Understanding that this is a fixed size window and that the tokens within this window work together to produce the output of the model is important when utilizing LLMs in a responsible manner. This set of features is aimed addressing the CONTEXNEG misconception.

In real transformer-based LLMs, the context window is implemented through a self-attention mechanism, which is a function that takes in the previous tokens and calculates a weighted sum of them to produce the output [69]. This is a complex process, which we deemed to be too technical for the target audience of the tool. Thus, we needed a viable representation of this attention mechanism that gives understanding on how this is a statistic evaluation, and that the other tokens in the context window work together to produce the output of the model.

In order to find a viable representation of the context window, we went back to the origins of NLP. In the mid 20th century, Claude Shannon proposed using Markov chains to model the statistical properties of language, this is what we now call n-gram models [23]. These models work by using token frequency counts to calculate the probability distribution of the n-th token given the previous $n-1$ tokens. By introducing this model to users, we can give intuition on how the context window works, and how the tokens within this window work together to produce the output of the model.

We provided users with three different context window features: unigram, digram and trigram. Each of the features was aimed at different aspects of the context window. The unigram feature starts of by showing we can use the frequency of individual tokens to predict the next token, without taking into account any previous tokens. The digram feature shows that looking at tokens in context may change the interpretation of them, and thus give a more varied output. Finally, the trigram model illustrates that by looking at a larger context window, the model will be able to produce more coherent and contextually relevant output.

Non-Determinism

An important aspect of LLMs is that they are non-deterministic, meaning that the same prompt can produce different outputs. The addition of this variability comes with the cost of unpredictability and also the risk of generated output not being true to the training data. Understanding the nature of how this randomness is introduced into the model is important for responsible use of LLMs. We added this feature to address the HALLBLIND misconception.

In the tool, this is implemented through a simple switch which toggles between a deterministic model, which always selects the most probable token as the output, and a non-deterministic model, which samples from the probability distribution of the next token. The sampling is done through a weighted random sampling, which means that tokens with higher probabilities are more likely to be selected as the output, but there is still a chance for less probable tokens to be selected. It is visualized in the tool by showing the probability distribution as a bar chart morphing into a pie chart, which then turn into a roulette wheel that spins and lands on the selected token.

A planned feature that was not implemented in the final version of the tool was a temperature slider, which would allow users to adjust the level of randomness in the model. This was planned to be implemented in a similar manner as it is in in modern LLMs, with a softmax function (4.1) that adjusts the probability distribution of the next token based on the temperature value.

$$\text{Softmax}(z_i) = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}} \tag{4.1}$$

Here z_i is the logit for token i, that is the the probability of token i being the next token before applying the softmax function. T is a temperature value that controls the level of randomness in the model. The plan was to set this value as a slider between 0.1 and 2, where a lower value ($T < 1.0$) would result in more deterministic behavior, while a higher value ($T > 1.0$) would flatten the probability distribution and make the model more random. A value of $T = 1.0$ would leave the probability distribution unchanged. This would have allowed users to explore how adjusting the level of randomness in the model can affect the output, and explain how this can be tweaked to a more convincing balance of accurate and creative output, depending on the use case.

Tokenization

The tokenization process converts input text into a format that the model can work with. This is done before it is fed into the model, and it is an important aspect of how LLMs work. LLMs needs a numerical representation of the input text in order to do the calculations needed to produce the output, and tokenization is the process of converting the input text into this numerical format. First by breaking the input text into smaller units called tokens, and then by converting these tokens into numerical representations in the form of word embeddings. We wanted the user to understand that LLMs do not understand, nor see, language in the same way as humans do and therefore should not be expected to respond to prompts in the same way as a human would. Understanding this will address the SEMUNDER misconception.

In the tool, the model used is not reliant on a numerical representation of the input text, but rather works directly with the tokens. Thus, we needed to find a viable representation of the tokenization process that gives users an understanding of how the input text is broken down into smaller units, and how these units are then used by the model to produce the output. The first step, is to understand that the entire input text is not processed as a whole, but broken down into tokens. The intuition given here is that the model already looks at the input text as a collection of words, but that this is not necessarily the best way to break down the input text for the model to work with. We use punctuation as an example of how the word-based tokenization can be problematic, as it will treat the same word as different tokens depending on the punctuation used with it. Thus, the first feature in this category will change from a whitespace-based tokenization to a punctuation-based Regular Expression (REGEX) tokenization.

The second step is to understand that the tokens are not just randomly selected based on the whitespace and punctuation in the input text, but that are designed to be somewhat language-aware. The syntax of natural language is complex and can be ambiguous, and this ambiguity must be captured in order for the model to be able to produce reliable output. We try to give users an intuition on this by using the Natural Language Toolkit (NLTK) library [6]. This library contains a collection of tools for working with natural language data, including a tokenizer that is designed to be language-aware. We discussed whether to include an additional feature focused on the design of word embeddings, discussing how this can include biases in the training data and how this can affect the output of the model. However, we did not find a viable way to represent this in the tool, and thus it was not included in the final version.

4.2 Study Design

4.2.1 Recruitment

Participants in the study were recruited through a Microsoft Forms survey, which was distributed through various channels. The survey was shared to students and teaching assistants in the CS1 course INF100 at UiB, as well as posted on forums of voluntary student organizations, and posters were put up around campus. To appeal to as many potential participants as possible, a lottery of cinema tickets was used as an incentive for participation. The recruitment targeted students with varying levels of technical background in order to capture a diverse range of understandings and misconceptions about LLMs.

The survey contained a brief description of the study, and participants were asked to provide their email address if they were interested in participating. The survey also contained a few questions about the participants' background including age group, occupation, and if they were a student, which institution they were affiliated with.

A total of 10 participants were recruited for the study, all students from different departments at UiB, with varying levels of technical background ranging from undergraduate students in informatics to undergraduate students in social sciences.

4.2.2 Assessment Interviews

The study started with an assessment interview, where participants were asked about their background and experience with LLMs, as well as their understanding of how LLMs work and their opinions on the technology. The interview was semi-structured, with a brief interview guide to ensure that relevant topics were covered, while still allowing for participants to express their thoughts and opinions in their own words. Follow-up questions were asked based on the participants' responses to explore if they contained specific misconceptions about LLMs. All interviews were conducted in Norwegian. The interview guide can be found in Norwegian in Appendix A. The interviews lasted between 15 and 45 minutes, with an average of around 30 minutes.

After the interviews were conducted, they were transcribed using the built-in transcription tool in Microsoft Word, and then manually checked for accuracy. The transcripts were analyzed to identify misconceptions about LLMs in the participants' understanding through a deductive content analysis approach, using the coding scheme in Appendix B.

4.2.3 Tool Usage

After the assessment interview, participants were given access to the mediation tool and were asked to use it during the upcoming week. They were not given any time requirement for how much they should use the tool, but were encouraged to explore it at their own pace and in their own time. The mediation tool was hosted on a virtual machine in the NREC cloud infrastructure service [56]. Participants were given a link to the tool, as well as a brief introduction to how to access and use it. They were also given contact information of the researcher in case they had any questions or issues with the tool.

4.2.4 Follow-up Interviews

Approximately one week after participants were given access to the mediation tool, they were invited to a follow-up interview. The purpose of this interview was to gather the participants' thoughts and opinions on the mediation tool, as well as to explore if they had any changes in their understanding of LLMs after using the tool. The interview was semi-structured, with a set of prepared questions to guide the conversation, but also allowing participants to freely express their thoughts and opinions. The interviews contained personalized questions based on the participants' responses in the assessment interview, as well as general questions about their experience with the tool. All interviews were conducted in Norwegian. The interview guide can be found in Norwegian in Appendix C. The interviews lasted between 10 and 20 minutes, with an average of around 15 minutes.

The follow-up interviews were transcribed and analyzed in the same way as the assessment interviews, with the transcripts being used for the analysis of the interviews.

4.2.5 Analysis

The interview transcripts were analyzed using deductive content analysis. A predefined coding scheme consisting of the 18 misconceptions identified in chapter 3 was used to code the transcripts. The coding sheet can be found in Appendix B. Each transcript was independently coded by two coders, where the presence or absence of each misconception category was noted. A transcript could be coded with multiple misconceptions. Coders could identify multiple excerpts corresponding to the same misconception within a transcript. However, for the reliability analysis, each misconception category was treated as a binary variable indicating whether the misconception was present or absent in the transcript. Each interview transcript was treated as a single coding unit.

After coding of the assessment interviews, the data was analyzed through IRR analysis to determine the level of agreement between the coders. The analysis was done in a Jupyter Notebook using Python 3.12. The IRR was calculated using percentage of agreement, Cohen's Kappa, calculated with the `cohen_kappa_score` function from the `sklearn.metrics` library [52], and Krippendorff's Alpha, calculated with the `krippendorff.alpha` function from the `krippendorff` library [8]. A IRR coefficient was calculated for each misconception category, treating the presence or absence of the misconception in each transcript as a binary variable. How the data is loaded and the IRR coefficients are calculated can be seen in Listing 4.1.

After calculating the IRR-scores, the two coders discussed any discrepancies in their coding and came to a consensus on the final coding for each transcript. The consensus reached on the coding of the assessment interviews were used as a baseline for the coding of the follow-up interviews. The IRR-analysis was done for the follow-up interviews in the same way as for the assessment interviews. The final coding consensus of the follow-up interviews was then used for the analysis to examine changes in the presence of misconceptions between the assessment and follow-up interviews. This was done by looking at the number of misconceptions that were marked as resolved, meaning that they were present in the assessment interview but not in the follow-up interview. The analysis was done both on an individual level, looking at each participant's change in misconceptions, and on a group level, looking at the overall change in misconceptions across all participants.

Listing 4.1: Code for loading coded data and computing IRR coefficients

```python
import numpy as np
import pandas as pd
import krippendorff
from sklearn.metrics import cohen_kappa_score

CSV_PATH = "data/coding.csv"
META_COLS = {"unit_id", "coder"}

# --- Load & align ---
df = pd.read_csv(CSV_PATH)
df.columns = df.columns.str.strip()
coder_a, coder_b = sorted(df["coder"].unique())
a = df[df["coder"] == coder_a].set_index("unit_id")
b = df[df["coder"] == coder_b].set_index("unit_id")
categories = [c for c in a.columns if c not in META_COLS]
items = sorted(set(a.index) & set(b.index))
a, b = a.loc[items, categories], b.loc[items, categories]

# --- Per-category metrics ---
rows = []
for cat in categories:
    v_a, v_b = a[cat].values.astype(float), b[cat].values.astype(float)
    mask = ~(np.isnan(v_a) | np.isnan(v_b))
    coding_table = np.array([v_a, v_b])
    coder_a_values, coder_b_values = v_a[mask].astype(int), v_b[mask].astype(int)
    if len(coder_a_values) == 0:
        continue
    prevalence = coder_a_values.mean()
    observed_values = np.unique(coding_table[~np.isnan(coding_table)])

    if len(observed_values) < 2:
        kappa = np.nan
        alpha = np.nan
    else:
        kappa = cohen_kappa_score(coder_a_values, coder_b_values)
        alpha = krippendorff.alpha(
            reliability_data=coding_table,
            level_of_measurement="nominal")

    rows.append({
        "Category":   cat,
        "Prevalence": round(prevalence, 3),
        "Cohen k":    round(kappa, 3) if not np.isnan(kappa) else np.nan,
        "Kripp a":    round(alpha, 3) if not np.isnan(alpha) else np.nan,
    })

metrics = pd.DataFrame(rows)
print(metrics.to_string(index=False))
```

Chapter 5

Results

5.1 Assessment Interviews

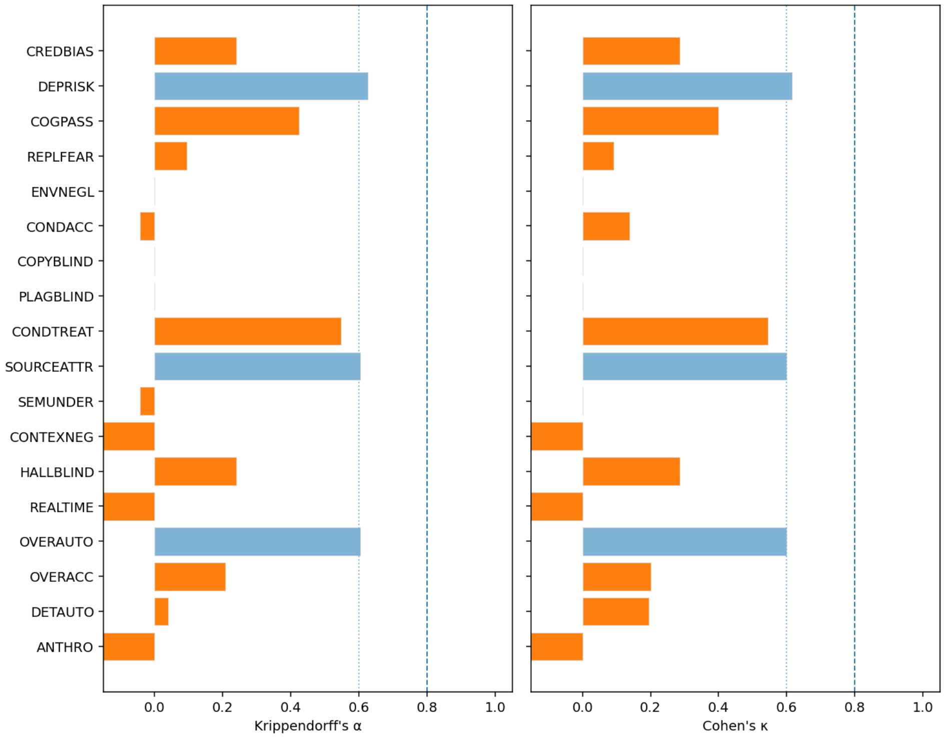
After the two coders completed their assessments, the codings were gathered for IRR analysis. Figure 5.1a shows the calculated agreement percentage, IRR-coefficients and prevalence across each category. ENVNEGL was not found in any of the transcripts by either coder, so a coefficient could not be calculated. Figure 5.1b shows the level of agreement on each category. We noticed a substantial agreement (both coefficients above 0.6) in the categories of COGPASS, SEMUNDER and OVERAUTO; a systematic disagreement (both coefficients below 0) in the categories of ANTHRO, REALTIME and CONTEXNEG; a minor systematic disagreement (one coefficient below 0) in the categories of SEMUNDER and CONDACC; and a minor agreement (both coefficients between 0 and 0.6) in the rest of the categories.

The coincidence matrices for each category are shown in Figure 5.2. Here, agreements are shown on the diagonal, while disagreements are shown on the antidiagonal.

There was no strong or perfect agreement on any of the categories, however the coefficients are vulnerable to a small dataset. The categories and codings were discussed and it was decided on a final coding matrix to be used for the follow-up interviews. The final coding is shown in Table 5.1.

Category	Agreement	κ	α	Prevalence
ANTHRO	50%	-0.19	-0.267	60%
DETAUTO	50%	0.194	0.040	30%
OVERACC	60%	0.200	0.208	30%
OVERAUTO	80%	0.600	0.604	30%
REALTIME	20%	-0.176	-0.520	80%
HALLBLIND	70%	0.286	0.240	10%
CONTEXNEG	50%	-0.316	-0.267	30%
SEMUNDER	50%	0.000	-0.044	50%
SOURCEATTR	80%	0.600	0.604	30%
CONDTREAT	80%	0.545	0.548	40%
PLAGBLIND	90%	0.000	0.000	10%
COPYBLIND	90%	0.000	0.000	10%
CONDACC	50%	0.138	-0.044	10%
ENVNEGL	100%	—	—	0%
REPLFEAR	60%	0.091	0.095	20%
COGPASS	70%	0.400	0.424	50%
DEPRISK	90%	0.615	0.627	10%
CREDBIAS	70%	0.286	0.240	40%

(a) IRR values for Assessment Interviews



(b) Agreement Coefficients for Assessment Interviews

Figure 5.1: IRR values and kappa/alpha plot for Assessment Interviews

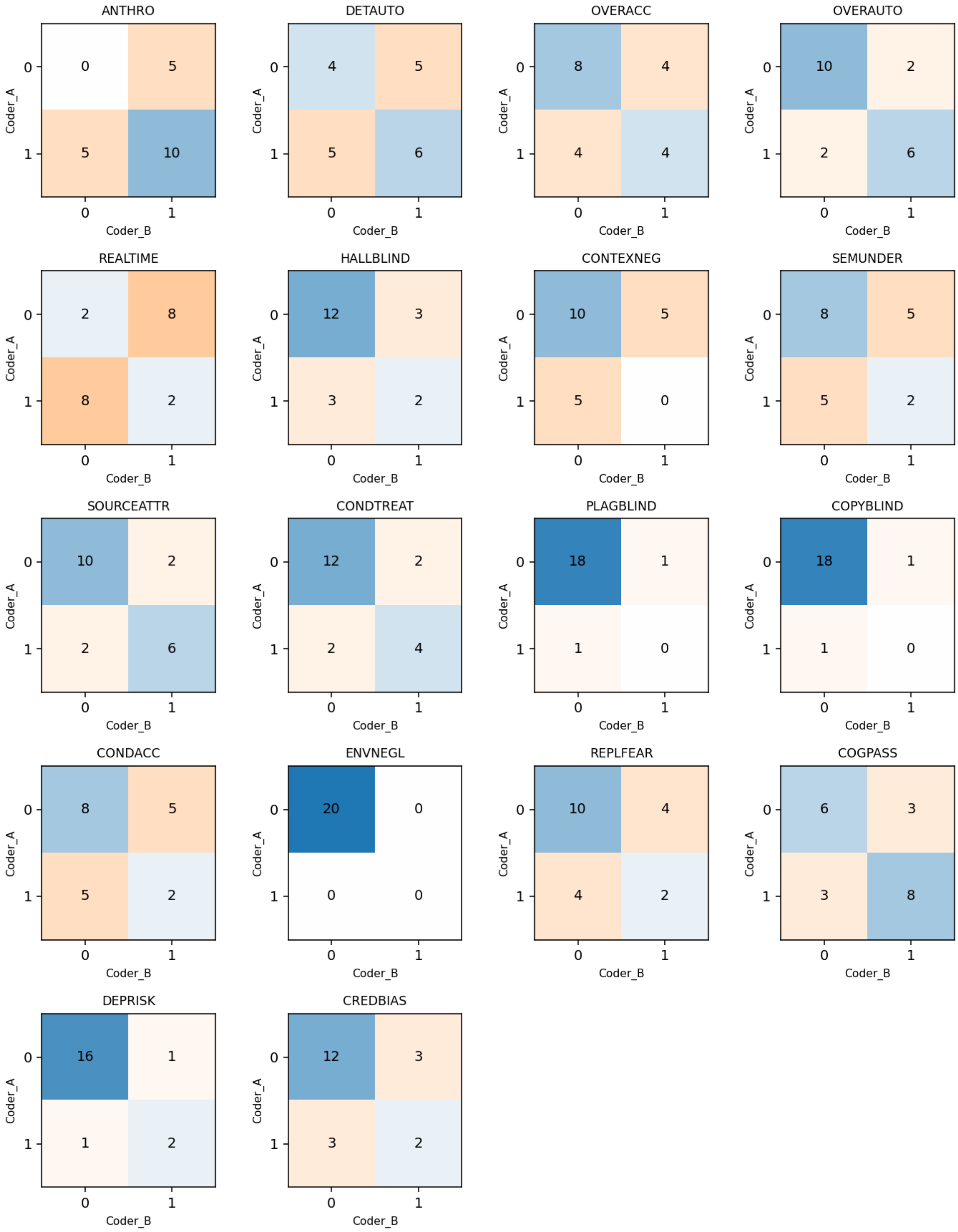


Figure 5.2: Coincidence Matrices for each Category in Assessment Interviews

(a) Dimension 1: FCA

p_id	ANTHRO	DETAUTO	OVERACC	OVERAUTO
1	1	0	1	1
2	1	1	0	0
3	1	1	0	0
4	0	1	0	0
5	1	0	1	0
6	1	0	0	0
7	1	0	1	0
8	1	1	1	1
9	1	0	0	0
10	0	0	0	1

(b) Dimension 2: MU

p_id	REALTIME	HALLBLIND	CONTEXNEG	SEMUNDER	SOURCEATTR	CONDTREAT
1	1	0	0	0	1	0
2	1	0	0	0	0	0
3	1	0	1	0	1	0
4	1	1	1	1	0	0
5	0	1	1	0	1	0
6	1	1	0	0	0	0
7	0	0	0	1	0	0
8	1	0	0	1	1	1
9	1	0	0	1	0	1
10	1	1	0	0	1	0

(c) Dimension 3: EAI

p_id	PLAGBLIND	COPYBLIND	CONDACC	ENVNEGL
1	0	0	0	0
2	0	0	0	0
3	0	0	1	0
4	0	0	0	0
5	0	0	1	0
6	0	0	0	0
7	0	0	1	0
8	0	0	0	0
9	0	0	0	0
10	0	0	1	0

(d) Dimension 4: SCI

p_id	REPLFEAR	COGPASS	DEPRISK	CREDBIAS
1	0	1	1	0
2	0	0	0	0
3	0	0	0	0
4	1	0	0	0
5	0	1	0	0
6	0	1	0	0
7	0	1	0	0
8	0	1	0	0
9	0	0	0	0
10	0	0	0	1

Table 5.1: Binary coding matrix for the assessment interviews, by dimension.

5.2 Follow Up Interviews

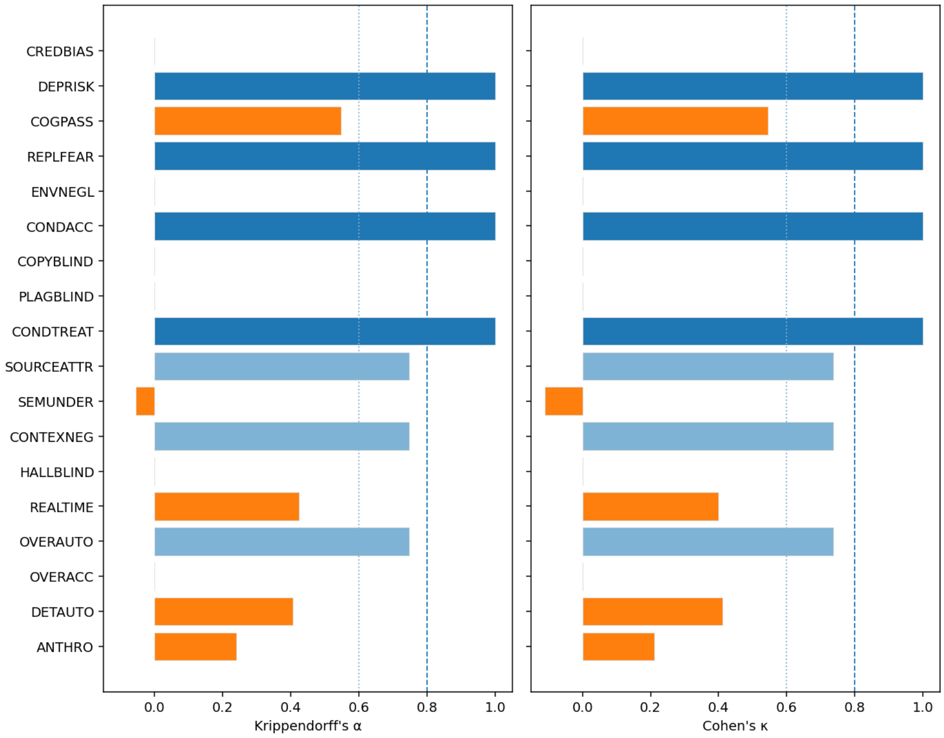
The final coding matrix from the assessment interviews, shown in Table 5.1, was used as a baseline for the coders when coding the follow-up interviews. Misconceptions deemed present in the assessment interview were flipped to absent if the participants actively made particular statements showing awareness or understanding of the correct concept. If they expressed the same opinion as they did in the assessment interview, or if they did not provide sufficient evidence to support awareness or viable understanding, the misconception was retained. Figure 5.3 shows the calculated IRR coefficients and agreement scores for the coders in the follow up interviews. From the assessment interviews, ENVNEGL, PLAGBLIND, and COPYBLIND were already absent, but in the follow-up coding CREDBIAS was found absent as well. Perfect agreement was found in the categories of CONDTREAT, CONDACC and REPLFEAR; and substantial agreement (both coefficients above 0.6) in the categories of OVERAUTO, CONTEXNEG and SOURCEATTR.

SEMUNDER was the only category with a systematic disagreement (both coefficients below 0). The rest of the categories had a minor agreement (both coefficients between 0 and 0.6). Coincidence matrices for each category are shown in Figure 5.4.

The coders met to discuss the coding of the follow-up interviews and agreed on a final coding matrix to be used for the analysis of the results. The final coding matrix is shown in Table 5.2.

Category	Agreement	κ	α	Prevalence
ANTHRO	70%	0.211	0.240	20%
DETAUTO	80%	0.412	0.406	30%
OVERACC	90%	0.000	0.000	10%
OVERAUTO	90%	0.737	0.747	30%
REALTIME	70%	0.400	0.424	50%
HALLBLIND	90%	0.000	0.000	0%
CONTEXNEG	90%	0.737	0.747	30%
SEMUNDER	80%	-0.111	-0.056	10%
SOURCEATTR	90%	0.737	0.747	20%
CONDTREAT	100%	1	1	20%
PLAGBLIND	100%	—	—	0%
COPYBLIND	100%	—	—	0%
CONDACC	100%	1	1	40%
ENVNEGL	100%	—	—	0%
REPLFEAR	100%	1	1	10%
COGPASS	80%	0.545	0.548	40%
DEPRISK	100%	1	1	10%
CREDBIAS	100%	—	—	0%

(a) IRR values for Follow Up Interviews



(b) Agreement Coefficients for Follow Up Interviews

Figure 5.3: IRR values and kappa/alpha plot for Follow Up Interviews

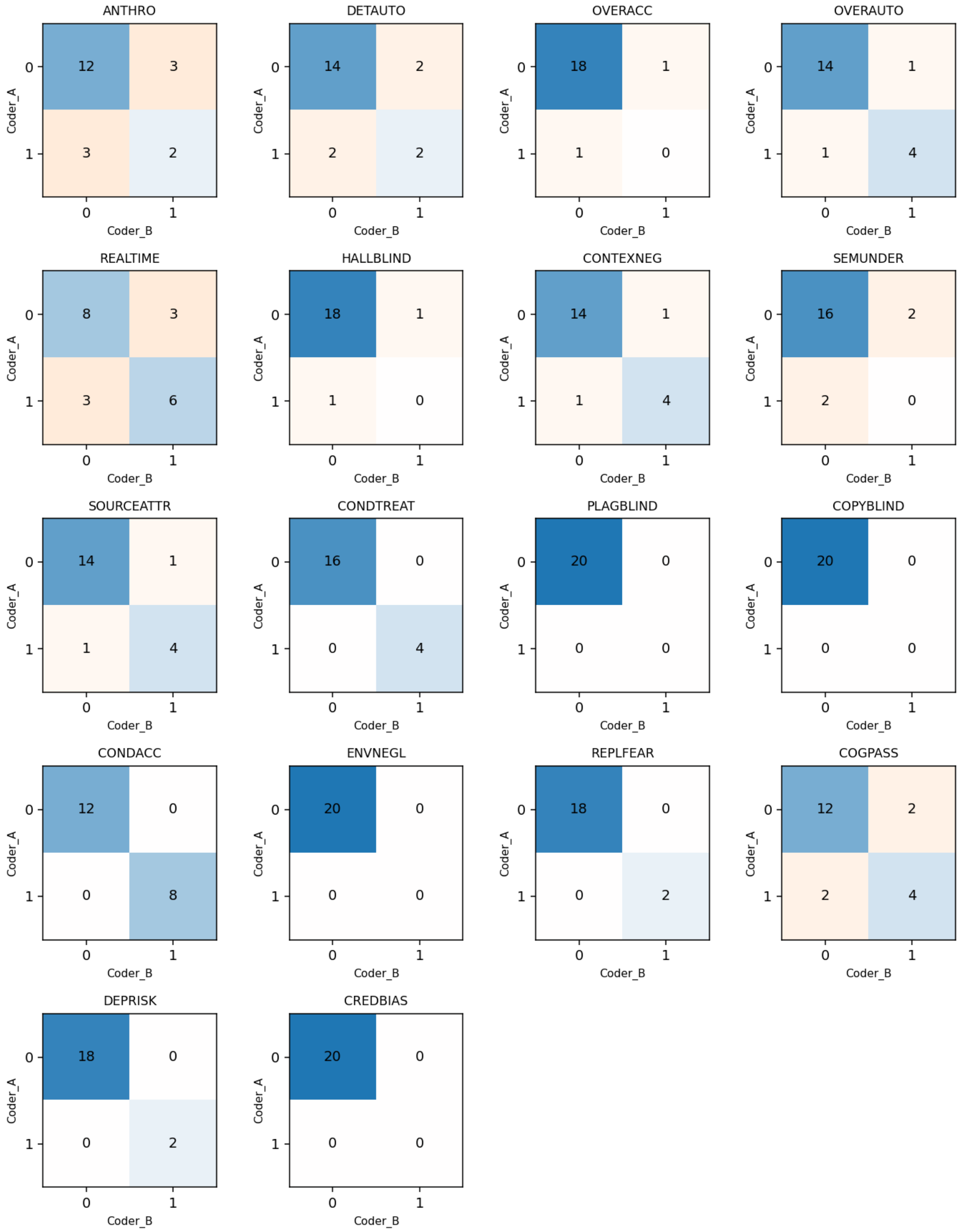


Figure 5.4: Coincidence Matrices for each Category in Follow Up Interviews

(a) Dimension 1: FCA

p_id	ANTHRO	DETAUTO	OVERACC	OVERAUTO
1	1	0	1	1
2	0	1	0	0
3	0	1	0	0
4	0	0	0	0
5	0	0	0	0
6	0	0	0	0
7	0	0	0	0
8	1	0	0	0
9	0	0	0	0
10	0	0	0	1

(b) Dimension 2: MU

p_id	REALTIME	HALLBLIND	CONTEXNEG	SEMUNDER	SOURCEATTR	CONDTREAT
1	1	0	0	0	1	0
2	0	0	0	0	0	0
3	0	0	1	0	1	0
4	1	0	0	0	0	0
5	0	0	1	0	1	0
6	1	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	1
9	0	0	0	0	0	1
10	1	0	0	0	0	0

(c) Dimension 3: EAI

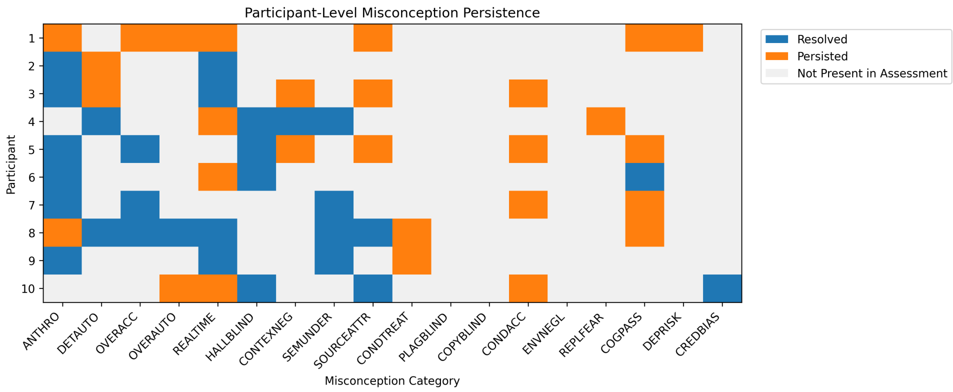
p_id	PLAGBLIND	COPYBLIND	CONDACC	ENVNEGL
1	0	0	0	0
2	0	0	0	0
3	0	0	1	0
4	0	0	0	0
5	0	0	1	0
6	0	0	0	0
7	0	0	1	0
8	0	0	0	0
9	0	0	0	0
10	0	0	1	0

(d) Dimension 4: SCI

p_id	REPLFEAR	COGPASS	DEPRISK	CREDBIAS
1	0	1	1	0
2	0	0	0	0
3	0	0	0	0
4	1	0	0	0
5	0	1	0	0
6	0	0	0	0
7	0	1	0	0
8	0	1	0	0
9	0	0	0	0
10	0	0	0	0

Table 5.2: Binary coding matrix for the follow-up interviews, by dimension.

Figure 5.5: Misconception Persistence Heatmap



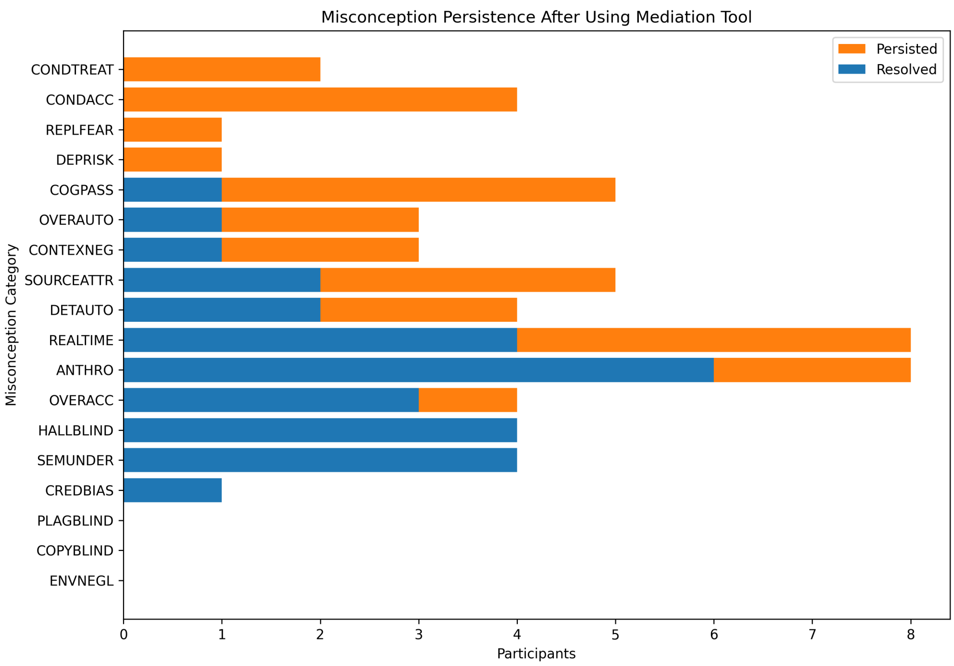
5.3 Shifts

When the final coding matrix for both the assessment (Table 5.1) and the follow-up interviews (Table 5.2) were compared, several shifts in coding were observed. A detailed heatmap of the shifts is shown in Figure 5.5, where a misconception is marked as resolved if it was present in the assessment but absent in the follow-up interviews.

To see any indication of a general effect of the mediation tool, we can look at the quantitative data on misconception persistence, shown in Figure 5.6. Notably, all participants in the categories of CREDBIAS, SEMUNDER, and HALLBLIND were found resolved; most participants in the categories of OVERACC and ANTHRO were found resolved; and about half of the participants in the categories of REALTIME, DETAUTO, and SOURCEATTR were found resolved. On the other hand, no participants were found resolved in the categories of CONDTREAT, CONDACC, REPLFEAR, and DEPRISK.

For the rest of the categories, the persistence was higher than the resolution. From these result we get an indication of a general effect of the mediation tool, which will be discussed in chapter 6 along with the interpretation of the results and the limitations of the study.

Figure 5.6: Quantitative data on Misconception Persistence



Chapter 6

Discussion

The following chapter interprets the results presented in chapter 5 in light of the research questions motivating this thesis. We begin by discussing the findings from the assessment and follow-up interviews, including the Inter-Rater Reliability (IRR) of the coding process and the observed shifts in misconceptions. We then discuss the effectiveness of the mediation tool in addressing the identified misconceptions, before reflecting on the methodological limitations of the study and directions for future work.

6.1 Interpretation of Results

The following section interprets the findings from both the assessment and follow-up interviews. We first discuss the inter-rater reliability of the coding process for each interview round, before turning to the co-occurrences of categories observed in the assessment interviews.

Assessment Interviews

After analyzing the results from the assessment interviews, several key insights emerged. Three out of four categories in the dimension of Ethical and Academic Integrity were found to be absent in the dataset. ENVNEGL was not found in any of the transcripts, by any of the coders, which indicates that the general public have been exposed to discussions about the environmental consequences of GenAI, and are aware of them to

Table 6.1: Cohen's κ and maximum achievable κ per misconception category (assessment interviews)

Category	κ	$\kappa_{\max}$
ANTHRO	-0.190	0.286
DETAUTO	0.194	0.194
OVERACC	0.200	0.600
OVERAUTO	0.600	0.600
REALTIME	-0.176	0.118
HALLBLIND	0.286	0.286
CONTEXNEG	-0.316	0.737
SEMUNDER	0.000	0.400
SOURCEATTR	0.600	0.600
CONDTREAT	0.545	0.545
PLAGBLIND	0.000	0.000
COPYBLIND	0.000	0.000
CONDACC	0.138	0.138
ENVNEGL	–	–
REPLFEAR	0.091	0.545
COGPASS	0.400	0.800
DEPRISK	0.615	0.615
CREDBIAS	0.286	0.286

some extent. Although most of the participants did not understand why these models are environmentally costly, they were aware of the fact that they are. PLAGBLIND and COPYBLIND were also absent in the dataset, but with initial disagreements between coders. Given the definitions of these categories in the taxonomy, the format of a semi-structured interview made it difficult to identify these categories. We do not think that these categories are necessarily absent in all participants, but rather that the format of the interview made it hard to notice them.

Given the small dataset of 10 participants, it was expected that the agreement coefficients would be vulnerable to small changes in the data. In an attempt of isolate the prevalence imbalance from coder disagreements, we compared the calculated κ with the maximum achievable $\kappa_{\max}$. Table 6.1 shows the calculated κ and $\kappa_{\max}$ for each category. Neglecting the non-observed categories, we see that DETAUTO, OVERAUTO, HALLBLIND, SOURCEATTR, CONDTREAT, CONDACC, DEPRISK and CREDBIAS all reach their theoretical maximum ($\kappa = \kappa_{\max}$), indicating that observed agreement is primarily constrained by the distribution rather than coder inconsistencies. Meanwhile, categories exhibiting $\kappa < \kappa_{\max}$ may indicate that the coders had different interpretations of the given definitions.

The categories with negative IRR coefficients are those where the two coders had different interpretations of the given definitions. It was a consensus between coders that the category of ANTHRO was such a broad concept that it was difficult to apply consistently. The disagreements here were mostly due to overlooking a specific phrasing or quote in the coding process. The definition of REALTIME was given as “Belief that the LLM retrieves current information.” Here, the coder disagreements came from what counts as “current information.” It was settled that this should count as both retrieving and/or utilizing information from the internet, the current conversation or other conversations as the interaction happened. For instance, this quote was marked by just one of the coders, but was in the end determined to be presence of the misconception:

> “It gets [information] on its server from conversations with other people continuously. And you can also say that one does not know how often the language model you are talking with gets new information, or not the language model, but the server you are talking with gets information from other servers.”[1]

Modern RAG-based LLMs have also made it more difficult to determine what counts as “retrieving current information”, as the line between retrieving and generating has become more blurred.

Another difference in interpretation was the definition of CONTEXNEG, which was defined as “Failure to recognize limits of LLMs’ context window.” One interpretation was that the “limits of the context window” should be recognized as the amount of tokens that can be processed at once, while the other interpretation was that it should be recognized as the fact that there is a context window and that limits what and which information the model can use to generate responses, as well as the limitations within that context window. It was settled that the second interpretation should be used, which led to cases like the following being marked as presence of the misconception:

> “Yes, it [the model] usually takes into account what I talked with it about earlier, both in conversations, but also outside of that conversation. It can pull in things from other conversations.” [2]

[1] Original in Norwegian: “Den får på serveren sin fra samtaler med andre mennesker kontinuerlig. Og så kan du også si at man vet ikke hvor ofte språkmodellen du prater med får inn ny informasjon eller ikke språkmodellen, men den serveren du prater med får informasjon fra andre servere.”

[2] Original in Norwegian: “Ja, den bruker å ta hensyn til hva jeg snakket med den om tidligere, både i samtaler, men også utenfor den samtalen. Den kan dra inn ting fra andre samtaler. ”

The remaining disagreements were not as systematic as the ones mentioned above, but were rather due to overlooking a specific phrasing or interpreting a specific phrasing as not sufficient to mark the category as present. These were quickly resolved by going through the disagreements together and discussing the specific cases to land on the consensus shown in Table 5.1.

Follow Up Interviews

The higher values of the IRR-coefficients on the follow-up interviews are expected as the consensus reached in the first assessment was used as a baseline when coding the follow-up transcripts. Therefore, there shouldn't be put too much emphasis on all of these values, other than the ones were there was observed disagreements. SEMUNDER was the only category with negative IRR-coefficients, this was marked preserved once by each of the coders on different participants. Both of these were declared resolved in the consensus, but were not picked up by both coders due to some ambiguity as to whether the given statements were sufficient to show proper understanding. For instance this response to the question of it is suitable to use LLMs for educational purposes:

> "I think, if you don't have anyone else you can ask, then I think it can work. But I've kind of come to understand that it doesn't really comprehend what it reads, so then it becomes difficult to know whether you're learning correctly." [3]

There was also some discussion regarding the category of ANTHRO. In the initial assessment interviews, this category was marked as present if the participant explicitly expressed anthropomorphic views about the model. The other categories were marked as resolved if the participant expressed a understanding contradicting the misconception, but for this category, it was not as clear what would count as a resolved misconception. It was settled that the category should be marked as resolved if the participant expressed a mechanistic understanding of the model, unlike the anthropomorphic views expressed in the assessment interviews. For instance, this quote was marked as resolved:

> "It just does coincidences, and it doesn't really know anything." [4]

Other than these two categories, the disagreements were again mostly due to overlooking a specific phrasing or interpreting a specific phrasing as not sufficient to mark the category as resolved.

[3]Original in Norwegian: "Jeg tror, hvis du ikke har noen andre du kan spørre, så tror jeg det [kan fungere]. Men jeg har på en måte forstått at den ikke helt skjønner hva den leser, så da blir det vanskelig å vite om du lærer riktig."

[4]Original in Norwegian: "Den gjør bare tilfeldigheter, og den vet egentlig ingenting."

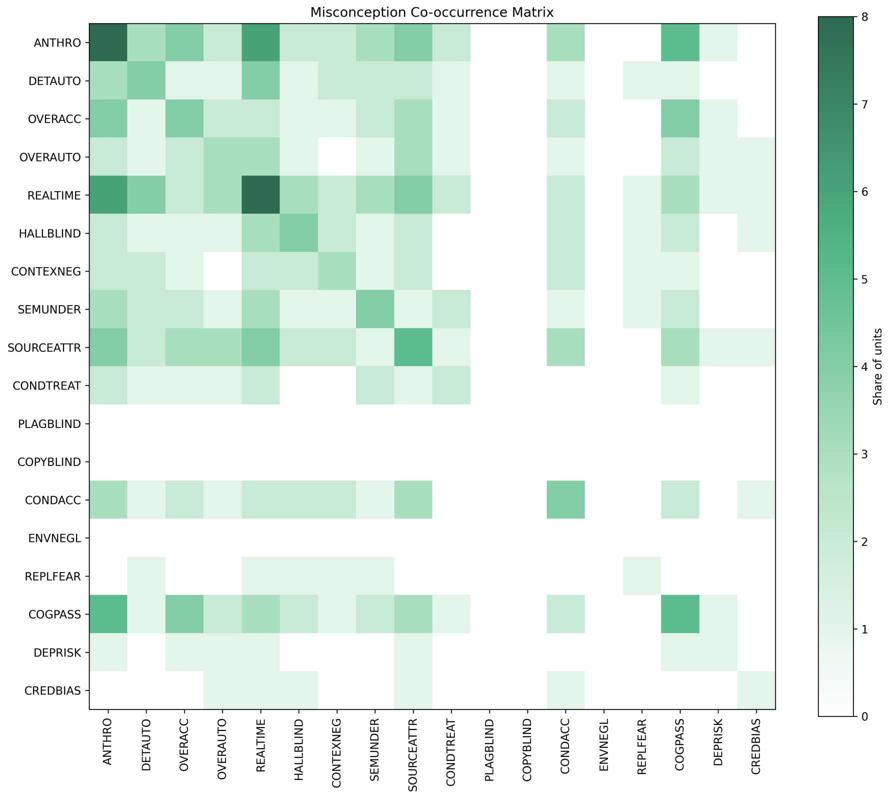


Figure 6.1: Co-occurrences of categories in the assessment interviews.

Co-Occurrences of Categories

In an attempt to identify potential relationships between the different categories, we looked at the co-occurrences of categories in the assessment interviews. A co-occurrence matrix is shown in Figure 6.1, where the color intensity of each cell represents the number of participants that had both categories marked as present in the assessment interview. The diagonal of the matrix can be interpreted as the prevalence of each category, which gives an indicator of how high the color intensity should be expected to be in the off-diagonal cells.

Again, the small sample size makes it difficult to draw any generalizable conclusions, but we can observe some patterns in the co-occurrences. All of the participants placed in the category of COGPASS also had the category of ANTHRO marked as present. This could indicate that users are more likely to do cognitive offloading to the model if they have anthropomorphic views of it.

It also shows that all participants placed in the category of OVERACC in the assessment interview, were also placed in the category of COGPASS, which indicates that users who overestimate the accuracy the reliability and/or correctness of LLM outputs are more likely to do cognitive offloading to the model. This is also an expected result, as users who overestimate the accuracy of the model are more likely to trust it and rely on it for tasks that they would normally do themselves. There is also a similar relation between OVERACC and ANTHRO, which indicates that users who have anthropomorphic views on the model might also be more likely to overestimate its capabilities, and therefore also more likely to do cognitive offloading to the model.

There is not sufficient evidence to draw any conclusions regarding other co-occurrences, but there is a potential relation between REALTIME and ANTHRO, which does make intuitive sense, as humans do operate with real-time information, and therefore users with anthropomorphic views on the model might also be more likely to believe that it operates with real-time information.

6.2 Mediation Tool Effectiveness

The final results of the follow-up interviews indicate that the mediation tool was effective in resolving some of the misconceptions, but not all. Most notably, 6 out of 8 participants who had the misconception of ANTHRO in the assessment interview, showed sufficient understanding of the mechanistic nature of the model in the follow-up interview, which indicates that the mediation tool was effective in resolving this misconception. This is a logical result, as the mediation gives a thorough explanation of the processes behind the model[5], which demystifies the model and makes it easier to understand that it is not a human-like entity. The other categories in the dimension of fundamental conceptual accuracy were only partially resolved, which might indicate that the gap between the simple model used in the mediation and modern LLMs is still too big to fully resolve these misconceptions.

The design of the mediation tool was to address the misconceptions surrounding the mechanistic nature of the model, therefore misconception in the dimension of mechanistic understanding holds the best picture in the effectiveness of the explanations given in the mediation. The fully resolved misconceptions of HALLBLIND and SEMUNDER indicate that the explanations of the tokenization process and the non-deterministic nature of the model were effective in resolving these misconceptions. The partially resolved misconceptions of REALTIME and SOURCEATTR indicate that the explanations of how training

[5] or at least simplified

data work could be improved for clarity about training as a process that happens before the model is deployed, and that the model does not have access to any information outside of the training data and the current conversation. To demonstrate that the model may use tokens from a variety of different sources, color-coded boxes were used in the animation, where different sources were represented by different colors. This might also suggest that it is possible to track the source of the tokens used by the model, which could be a potential source of confusion for users. This could be made clear by explicitly stating that the model does not know where the tokens come from or add a possibility to have more than two books available as training data at the same time. REALTIME was resolved in 4 out of the 8 cases where it was present in the assessment interview, but this might not be a result of the explanations given in the mediation, but rather a result of the participants exploration of the model and the realization that it does not have access to any information outside of the training data. This is demonstrated by the following quote:

> "I tried to get it to tell a story with a cat, but it really just wanted to talk about whales…" [6]

Through experimentation, the participant appeared to recognize that the model generates responses based on patterns learned during training rather than on access to external or real-time information.

Only 1 out of 3 cases of CONTEXNEG were resolved, which indicates that the n-gram model alone is not sufficient to give enough understanding regarding the limitations of the context window. The gap between the simple n-gram model used in the mediation and modern LLMs might be too big to fully resolve this misconception, as modern LLMs have a much larger context window.

An interesting observation is that one of the participants showed indication of a misconception not covered by the taxonomy as a result of using the mediation tool. It was found in the following quote when asked which of the features the participant had learned existed in a LLM gave the impression of storing information given:

> "I tried giving it the same prompt several times. There were new suggestions each time, both with the same texts and with different texts. I got different

[6] Original in Norwegian: "Jeg prøvde å få den til å fortelle en historie med katt, og den ville jo egentlig bare snakke om hvaler…"

> words every time I entered one, then deleted it, and then tried entering it again." [7]

Here, the participant seems to confuse what information lies in the context window and what information lies in the training data, which indicates a potential misconception regarding the different sources of information for the model. In the mediation tool, the users are able to select up to two books from a selection of nine books, this is the only module without further explanation. It is possible that this way of selecting the training data might be confused with how modern RAG-based LLMs let users upload files into the context window. This particular participant was also the only one who expressed that they were less skeptic about the technology after using the mediation tool, which might come as a result of them believing that they can influence the training data by using this uploading feature of modern RAG-based LLMs.

The mediation tool showed little effect in the other dimensions of Ethical and Academic Integrity and Socio-Cognitive Impact, indicating that mechanistic understanding alone is not sufficient to resolve misconceptions in these dimensions. When asked if using the mediation tool would in any way change the way they use GenAI, most of the participants said that it would not change how much they use it, but rather how they think about the responses generated by the model. Some also stated that they would be more careful when using LLMs for tasks that require semantic understanding, but that they still saw the strengths of the model in other tasks, such as brainstorming and formulating ideas.

When asked how much time they had spent using the mediation tool, answers varied from half an hour to four hours. Only one participant had none of their misconceptions marked resolved, this participant was among the ones who had spent the least amount of time using the mediation tool.

6.3 Methodological Limitations

As is typical of qualitative research, generalizability was not the primary aim of this study, but rather to get an indication of the prevalence of misconceptions about LLMs among

[7] Original in Norwegian: "Jeg prøvde å gi den samme prompten flere ganger. Det var nye forslag hver gang, både med samme tekster og med forskjellige tekster. Så fikk jeg ulike ord hver gang jeg la inn ett, og så slettet jeg det, og så prøvde jeg å legge inn det igjen."

users, and to get an indication of the effectiveness of the mediation tool in addressing these misconceptions. The selection of participants was done through convenience sampling, which might have led to a sample that is not representative of the general population of LLM users. However, no specific demographic was targeted, and the sample did include students in a variety of different fields and with different levels of experience with GenAI. The semi-structured interview format was chosen to allow for flexibility in the interviews, but this also made it difficult to identify certain categories in the taxonomy, such as PLAGBLIND and COPYBLIND. A more structured interview format, with specific questions aimed at identifying these categories, might have been more effective in identifying the presence of these misconceptions.

Optimally also, the analysis of the assessment interviews could have been completed before starting the process of the follow-up interviews, as this would enhance the questions asked during that second interview. Due to time limitations it was not possible to do the final coding on the assessment interviews before the follow-up interviews started. With the final coding available, it would be easier to ask tailored questions aimed at the specific categories market for that participant. Nevertheless, my own involvement in the coding process of both the assessment and follow-up interviews did mean that I had my preliminary notes available and I was able to tailor the follow-up interviews to some extent, mitigating this limitation partially.

The website itself may also have some limitations in its design that could have affected the results. The general feedback from the participants was however positive.

6.4 Future Work

The following section outlines directions for future work based on the findings and limitations of this study. It firstly involves potential improvements to the mediation tool itself.

Tool Improvements

Several areas of improvement were identified through the user testing and follow-up interviews. This includes room for expansion, but also some technical improvements to the current design of the tool.

Technical Feedback

There were some technical feedback regarding the mediation tool that could be improved in future iterations. Some users struggled with understanding the use of the prompt bar and the skill-tree view. The current solution works by prompting the model only if the last symbol in the prompt bar is a space. This is stated in the tutorial, but some users overlooked this. A potential solution to this could be to prompt the model whenever the user stops typing for a certain amount of time, and add a space between the user input and the model output if this is not the case already. Another solution could also be to prompt the model at every keystroke, but this would require a more efficient implementation of model prompting.

There was also some feedback regarding the corpus selection feature. This feature did not contain any explanation other than a menu where you were told that you could select up to two books to add to the training data. This might have caused some confusion regarding how this feature works, and how it relates to the interactions of modern RAG-based LLMs. Some participants also expressed that they wanted more variety in the selection of training data, this would be a good improvement for future iterations of the tool, as it would allow users to explore how different training data can affect the output of the model in a more diverse way. This could include different types of text, but also different languages, as the current selection of books are all in English. The original plan was for the user to be able to upload their own files as training data, but this was not implemented as the current solution for the different model configurations requires the model to be pre-trained.

Other technical feedback included some minor fixes. For instance, some users expressed issues with the scrollytelling chapters, where the text boxes at certain moments covered the animations, or that the explanations used terms that were not yet explained in the tool. There is also no support for mobile devices at the moment, which the majority of the users expressed that they would like to have. Users also found it annoying that adding a feature back into their configuration would send them to the scrollytelling chapter for that feature, even if they had already been through that chapter.

Expansion

A common theme in the follow-up interviews was that it was difficult to connect the simple n-gram model used in the mediation tool to the more complex modern LLMs. This gap might hinder the effectiveness of the mediation tool, as users might struggle to see how the explanations given in the mediation relate to the models they use in real life. Therefore, an expansion of the mediation tool to include more complex models could be beneficial in improving the effectiveness of the mediation. This could include adding more features to the skill-tree, such as the planned temperature slider and a introductory chapter for word embeddings. After these features are added, further explanation would need a re-design of the model. This can be solved by unlocking a new skill-tree and builder page once the user has added every feature in the current skill-tree. With this structure, the models can gradually increase in complexity as the user progresses through the mediation, until it reaches a level of complexity that is more similar to modern LLMs. This would reduce the gap between the mediation and the models that users interact with in real life, while still maintaining viable mental models for users to be able to not complete the course in its entirety.

Taxonomy modifications

The mediation tool is not the only part of this study which could be improved, as the taxonomy itself might be refined. The possible misconception which was discovered in the follow-up interviews regarding the confusion between the context window and the training data, indicates that there might be a need to add a category to the taxonomy regarding misconceptions about the different sources of information for the model. This should be added in the dimension of mechanistic understanding, as it is a misconception regarding the underlying processes of the model.

Not much data was found in the dimensions of Ethical and Academic Integrity and Socio-Cognitive Impact, which makes it difficult to draw any conclusions regarding the prevalence of misconceptions in these dimensions. This is more likely to be a result of the format of the interviews, which were more focused in general on the two dimensions of mechanistic understanding and fundamental conceptual accuracy. It is possible that more data in these dimensions could be found by using a different format for the interviews, where the focus is more on how users think about the ethical and societal implications of their own use of GenAI. A potential study could involve participants taking a stance on a specific dilemmas regarding the use of GenAI.

Due to the disagreements between coders regarding the definitions of ANTHRO, REALTIME and CONTEXNEG, it might be worth revisiting the definitions of these categories to make them more clear and less open to interpretation. The addition of a potential new category could also alter the definition of both REALTIME and CONTEXNEG.

Further research

With the improvements and modifications mentioned above, further research could be done to find more data on the prevalence of misconceptions in the different dimensions, and to further test the effectiveness of the mediation tool. With a larger sample size, it would also be possible to do more statistical analysis on the co-occurrences of categories, which could give more insight into potential relationships between the different misconceptions. I think it would be beneficial if a reiteration of the study were focused solely on the first two dimensions, as the scope might be too broad to find sufficient data in all four dimensions.

It would also be interesting to do somewhat of a reverse-minded study regarding the effectiveness of the mediation tool, where participants would be categorized into viable mental models instead of misconceptions, and the mediation tool would be tested on its ability to help users reach a viable mental model. In this study, although some misconceptions were marked as resolved, it is not clear whether the participants reached a viable mental model of the technology.

The mainstream LLMs are rapidly evolving, and new features and installments are being marketed as constant improvements to the technology. It would be interesting to do a study on the misconceptions surrounding these new features, such as RAG, CoT or reasoning models. These features do not alter the underlying processes of the model, but they do alter the responses generated by the model, leading to a different user experience, which could lead to new misconceptions.

Chapter 7

Conclusion

With the industry's huge commitment to the development of GenAI and LLMs, it is important to address the gap between the public adoption of the technology and the public understanding of the technology. This thesis has proposed a taxonomy of misconceptions about LLMs, and has developed a mediation tool aimed at addressing these misconceptions. The results of the study suggest that the mediation tool can be effective in addressing some misconceptions.

As to whether sufficient education has accompanied public adoption, the results suggest that there is still a notable prevalence of misconceptions about LLMs among users. This primarily includes anthropomorphization of the technology, and a lack of understanding of the underlying mechanisms of tokenization, training data and the context window. Some aspects surrounding LLMs are already well communicated, like the environmental impact, but not to a degree that appears to meaningfully influence how users engage with the technology. The taxonomy proposed in this thesis can be a useful tool for identifying misconceptions for further research and for the development of educational materials.

On addressing the misconceptions, the interactive learning experience provided by the mediation tool developed in this thesis shows promising results in helping users understand some inner workings of LLMs, resolving misconceptions such as anthropomorphization, hallucination blindness and belief of semantic understanding of tokens. However, the gap between the mediation and the complexity of modern LLMs appears to be too large to affect the participants use of the technology in a significant way.

Future work could involve expanding the mediation tool to bridge this gap, and to further test its effectiveness in addressing misconceptions about LLMs. The taxonomy can also be further refined based on the results of this study, and can be used as a basis for further research on the topic.

As LLMs continue to become further embedded in work, education and daily life, it is important to continue researching the misconceptions surrounding the technology. By identifying and addressing these misconceptions, we can help ensure that users are able to use the technology in a responsible and effective manner. Thus, focus should be shifted from the question of "Can machines think?" to the question of "How do *we* think about machines?"

List of Acronyms and Abbreviations

AI Artificial Intelligence.

API Application Programming Interface.

CoT Chain of Thought.

CS Computer Science.

GenAI Generative Artificial Intelligence.

GPT Generative Pre-trained Transformer.

GUI Graphical User Interface.

HAI Human-AI Interaction.

IRR Inter-Rater Reliability.

LLM Large Language Model.

LLMs Large Language Models.

NLP Natural Language Processing.

NLTK Natural Language Toolkit.

RAG Retrieval-Augmented Generation.

REGEX Regular Expression.

UiB The University of Bergen.

ZPD Zone of Proximal Development.

ZPF Zones of Proximal Flow.

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Appendix A

Assessment Interview Guide

This appendix contains the interview guide used for the assessment interviews, which were conducted to classify misconceptions about LLMs among the participants.

Spørsmål til Forhåndsintervju, LLM-explainer

Forklar med egne ord hva AI er og hvordan det fungerer

- Etter dette spørmålet skal AI i denne sammenhengen være forstått som språkmodeller, og ikke AI generelt.
- Dersom det er naturlig, still konkrete oppfølgingsspørmål for å dekke de ulike kategoriene:
  - Hvordan opprerer språkmodeller? (OVERAUTO, REALTIME, CONTEXNEG, SEMUNDER, SOURCEATTR, CONDTREAT)
  - Til hvilken grad stoler du på en språkmodells forståelse og svar? (OVERACC, HALLBLIND, SEMUNDER)
  - Hvor kommer informasjonen fra når du bruker en språkmodell, er det noen problemstillinger rundt dette og hvem som eier denne informasjonen? (SOURCEATTR, PLAGBLIND, COPYBLIND)

Forklar hvordan du bruker språkmodeller i hverdagen, og hva du bruker dem til

- Bruk bruksområdene som blir nevnt til å spørre om hvordan denne prosessen fungerer, og hva de tror skjer "bak kulissene" når de bruker språkmodeller.

- Spør om en spesifik prompt de kunne brukt
- Prøv også å finne ut av til hvilken grad de stoler på språkmodeller, og hva de gjør for å verifisere informasjonen de får fra dem. Også om det er noen situasjoner der de ikke stoler på språkmodeller, og hvorfor.

Hvordan påvirker språkmodeller deg i hverdagen, både positivt og negativt?

- Få vedkommende til å drøfte hvilke effekter dette har

Hvordan påvirker språkmodeller samfunnet generelt, både nå og i fremtiden?

- Få vedkommende til å drøfte hvilke effekter dette har, både på kort og lang sikt, og både positivt og negativt.

Appendix B

Coding Sheet

This appendix contains the coding sheet used for the deductive content analysis of the interviews. The coding sheet is based on the taxonomy of misconceptions presented in chapter 3, and contains definitions and examples for each category in the taxonomy. The coding sheet was used by the coders to identify and classify misconceptions in the interview transcripts.

Dimension 1: Foundational Conceptual Accuracy

Anthropomorphism

Attributing human mental states, intentions or understanding to a LLM.

This is a quite broad category, but subjects should be placed in this category if they in some way expressed belief in attributes not fitting of a statistical model and such attributes are aligned with human mental states.

Deterministic Automation

Confusing LLMs with deterministic automation systems.

This could be deterministic chatbot systems, but also more traditional automation systems such as a calculator, or algorithms in general. Subjects who in some way expressed belief that LLMs operate based on deterministic rules, or that they are deterministic systems should be categorized into this category.

Accuracy Overestimation

Overestimating the reliability and correctness of LLM outputs.

Subjects who in some way expressed belief that LLMs are more reliable, correct or trustworthy than they actually are should be categorized into this category.

Autonomy Overestimation

Belief that the LLM operate independently or possess agency.

Subjects who in some way expressed belief that LLMs operate independently, or possess agency should be categorized into this category. This could be belief in some sort of action from the model other than generating output, like “searching the internet”.

Dimension 2: Mechanistic Understanding

Real-time Data Belief

Belief that the LLM retrieves current information.

Subjects who in some way expressed belief that LLMs retrieve current information should be categorized into this category. This retrieval can be from the internet, from some database, or from the prompt itself.

Hallucination Blindness

Failure to recognize that LLMs can produce false or fabricated information.

This is not just about recognizing that LLMs can produce false information, but also recognizing that hallucinations is part of the core idea of how LLMs work, and that it is not just a bug or an error that can be fixed.

Context Window Neglect

Failure to recognize limits of LLMs' context window.

If subjects shows no indication of being aware of the limits of LLMs' context window should be categorized into this category. This could be expressed as belief that the model can keep track of an entire conversation, or that it can remember information from previous conversations.

Semantic Understanding

Lack of awareness that the LLM processes language as tokens rather than semantic units.

Subjects who in some way expressed belief that LLMs understand language on a semantic level, or that they read the text as given to them.

Source Attribution Error

Belief that LLM output originate from specific identifiable sources.

Subjects who in some way expressed belief that LLM output originate from specific identifiable sources should be categorized into this category.

Conditional Treatment

Belief that LLMs process prompts differently based on the content or context.

Subjects who in some way expressed belief that LLMs will process one prompt differently than another should be categorized into this category.

Dimension 3: Ethical and Academic Integrity

Plagiarism Blindness

Failure to recognize ethical and legal issues related to the traceability of sources behind LLM-generated content.

This is strongly correlated to the source attribution error, but is more focused on the ethical and legal issues related to the issue. A LLM will mimmic citation patterns in the training data, and failure to recognize the ethical and legal issues related to the traceability of sources behind LLM-generated content should be categorized into this category.

Copyright Blindness

Failure to recognize that LLM-generated content may still be subject to copyright and usage restrictions, despite being machine-generated.

This category is focused on the subjects ability to recognize the questions surrounding ownership of LLM-generated content. That is wether it can be claimed by the prompt creator, the LLM creator, or the original creator of the content in the training data. Failure to recognize this issue should be categorized into this category.

Conditional Acceptance

Ethical acceptance of LLM use only under certain conditions or domains.

Subjects who in some way expressed that "use of LLMs is fine if used for . . . " or "LLM use is not fine in my own field, but in . . . " should be categorized into this category.

Environmental Neglect

Failure to recognize environmental impact of LLMs.

Subjects who in some way expressed belief that LLMs have negligible environmental impact, or showed lack of awareness of energy consumption associated with LLM training and operation should be categorized into this category.

Dimension 4: Socio-Cognitive Impact

Replacement Fear

Belief that LLMs will replace human roles or professions, expertise, or professions.

Subjects who in some way expressed belief that LLMs will replace human roles or professions should be categorized into this category. Subjects should not be categorized into this category if they only expressed belief that LLMs will replace some tasks within a profession.

Cognitive Passivity

Tendency to rely on LLMs rather than engaging in independent reasoning or critical thinking.

Subjects who, when they described how they use LLMs, showed a tendency to rely on LLMs for tasks that would normally require independent reasoning or critical thinking, should be categorized into this category.

Dependency Risk

Failure to recognize that extensive can lead to long term reliance on LLMs, skill degradation or loss of autonomy.

Subjects who, in describing their use of LLMs, showed no recognition of the risk of becoming reliant on LLMs or described tasks that should not require LLMs but still used them for such tasks, should be categorized into this category.

Credibility Bias

Treating LLM outputs as inherently truthful or authoritative without critical evaluation.

Subjects could think of LLM output as reliable just because of the fact that is machine generated. If this is shown in the subjects description of their use of LLMs, they should be categorized into this category.

Appendix C

Follow-Up Interview Guide

This appendix contains the interview guide used for the follow-up interviews, which were conducted to assess the effectiveness of the mediation tool in addressing misconceptions about LLMs among the participants.

Spørsmål til Oppfølgningsintervju

Hva synes du om verktøyet?

- Hvor lang tid brukte du på det?
- Hvor enkelt var det å bruke det?
- Hvilke funksjoner brukte du mest?
- Hva likte du best med det?
- Hva likte du minst med det?
- Hva kan forbedres med det?
- Hvilke eksterne ressurser brukte du?

Har du fått nye meninger om språkmodeller?

- Har forståelsen din av hvordan språkmodeller fungerer endret seg etter å ha brukt verktøyet?
- Hva tenker du nå om…?
  - Spør spesifikt om misforståelser de hadde før

Har du blitt mer eller mindre skeptisk til teknologien etter å ha brukt verktøyet?

Kommer du til å forandre bruk av språkmodeller etter å ha brukt verktøyet?

- I så fall, hvordan?

Appendix D

Statement on the use of AI tools

Microsoft Copilot was used several times during the making of this thesis. It was used for code completion in the data analysis process for small snippets of code, such as plot formatting and data loading. This was mainly done to save time as the heavy lifting of the data analysis is done by the libraries used. It was also used for LaTeX code completion when making tables, figures and formatting.

For certain short phrases and sentences, ChatGPT or Claude.ai was used for suggestions on clarity and grammar. This was done in order to improve the readability of the text. No text within the thesis was generated by AI, but rather suggestions for improvement were made based on the original text.

Elicit was used as a supplementary literature search tool, to find relevant papers. The AI summaries were not used, but the search function was used to find relevant papers.

Microsoft Word's built in transcription tool was used to transcribe the interviews, which were then manually checked for accuracy and corrected where necessary.

I am aware that I am responsible for all content of this master's thesis.